US012523855B2

(12) **United States Patent**
Dewald et al.

(10) **Patent No.: US 12,523,855 B2**
(45) **Date of Patent: Jan. 13, 2026**

(54) **ANAMORPHIC LENS ASSEMBLIES**

(71) Applicant: **Atlas Lens Co.**, Glendale, CA (US)

(72) Inventors: **Duane Scott Dewald**, Glendale, CA (US); **Forrest Tanner Schultz**, Glendale, CA (US)

(73) Assignee: **Atlas Lens Co.**, Glendale, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 345 days.

(21) Appl. No.: **18/114,210**

(22) Filed: **Feb. 24, 2023**

(65) **Prior Publication Data**

US 2024/0288671 A1 Aug. 29, 2024

(51) **Int. Cl.**
G02B 13/08 (2006.01)
G02B 9/04 (2006.01)
G02B 9/64 (2006.01)

(52) **U.S. Cl.**
CPC ***G02B 13/08*** (2013.01); ***G02B 9/04*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,048,284	A	7/1936	Newcomer	
2,890,622	A	6/1959	Walter	
2,940,372	A	6/1960	Ehrenhaft et al.	
3,410,630	A	11/1968	Jacobs	
3,682,533	A	8/1972	Vetter	
3,871,748	A	3/1975	Day	
5,465,178	A *	11/1995	Nakai	G02B 19/0014 359/671
7,654,678	B1	2/2010	Runco et al.	
10,126,533	B1 *	11/2018	Caldwell	G02B 9/64
10,942,357	B2	3/2021	Suhara	
10,969,568	B1	4/2021	Li et al.	
2005/0225876	A1	10/2005	Nurishi	
2006/0050403	A1	3/2006	Neil	
2007/0081257	A1	4/2007	Bowron et al.	
2010/0208149	A1	8/2010	Morikuni	

(Continued)

OTHER PUBLICATIONS

International Search Report and Written Opinion, PCT App. No. PCT/US23/29659, Nov. 27, 2023, 9 pages.

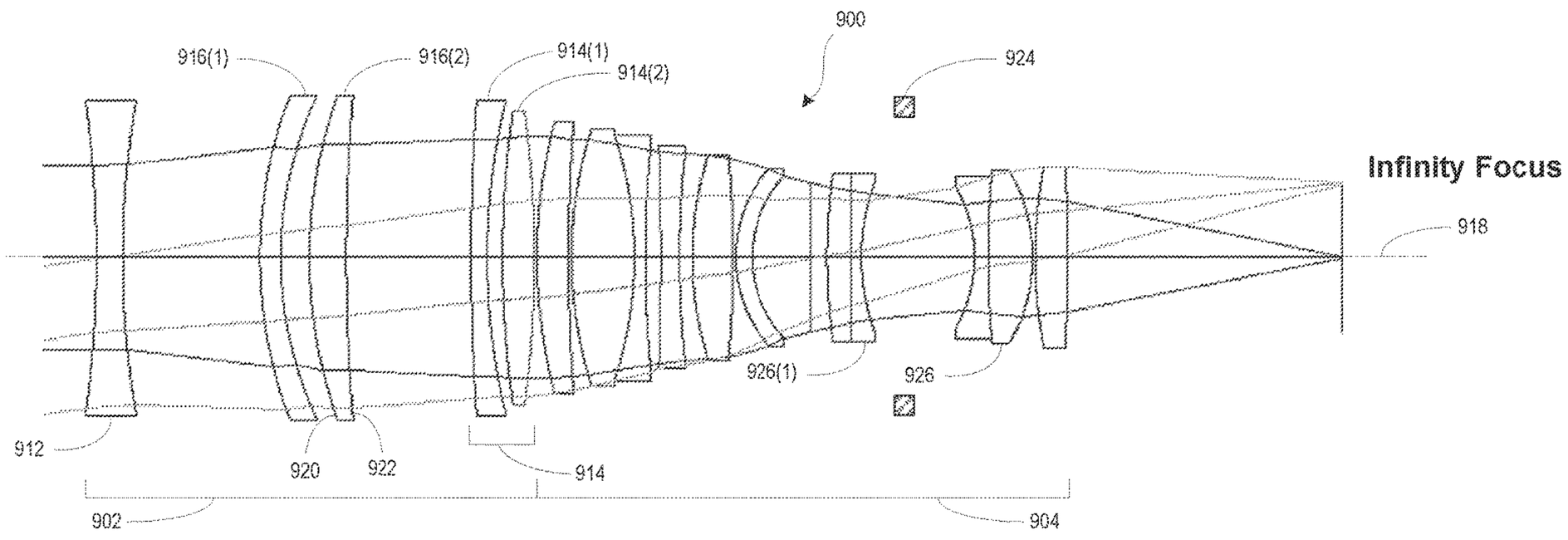
(Continued)

Primary Examiner — Darryl J Collins
(74) *Attorney, Agent, or Firm* — Nicholson De Vos Webster & Elliott LLP

(57) **ABSTRACT**

Lens assemblies according to the disclosure include an anamorphic lens component and a primary lens component. The anamorphic lens component includes cylindrical lens elements and one or more spherical lens elements disposed between the cylindrical lens elements and translatable along an optical axis of the anamorphic lens assembly with respect to the cylindrical lens elements. Optical characteristics of the anamorphic lens component are adjustable as the one or more spherical lens elements translate along the optical axis such that the cylindrical lens elements and the one or more spherical lens elements in combination produce a first astigmatism that is opposite to a second astigmatism produced by unequal movement of an image plane along horizontal and vertical axes as an object moves from infinity focus toward the lens assembly.

20 Claims, 12 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

2014/0016210	A1*	1/2014	Valles	G02B 13/18 359/671
2016/0170173	A1	6/2016	Macintosh et al.	
2016/0170186	A1	6/2016	Neil	
2017/0212334	A1	7/2017	Imaoka	
2018/0284401	A1	10/2018	Noethen	
2019/0094488	A1	3/2019	Macintosh et al.	
2020/0264412	A1	8/2020	Neil	

OTHER PUBLICATIONS

Non-Final Office Action, U.S. Appl. No. 17/899,448, Dec. 26, 2024, 17 pages.
Notice of Allowance, U.S. Appl. No. 17/899,448, Apr. 14, 2025, 11 pages.

* cited by examiner

FIG. 1A
Infinity Focus

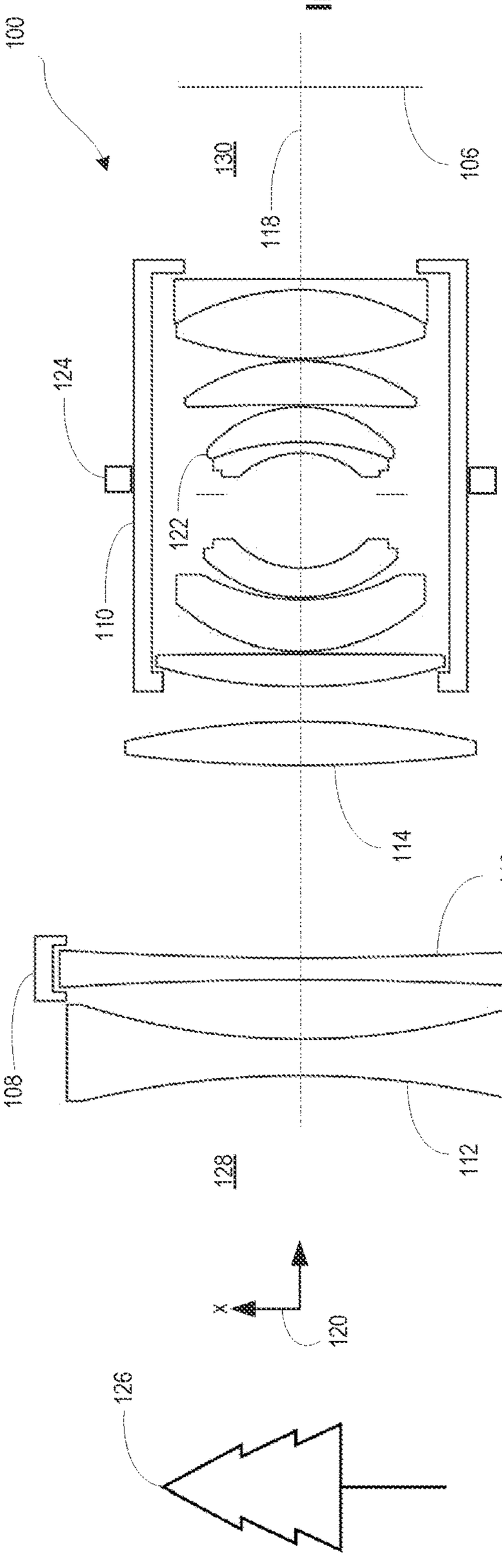


FIG. 1B
Close Focus

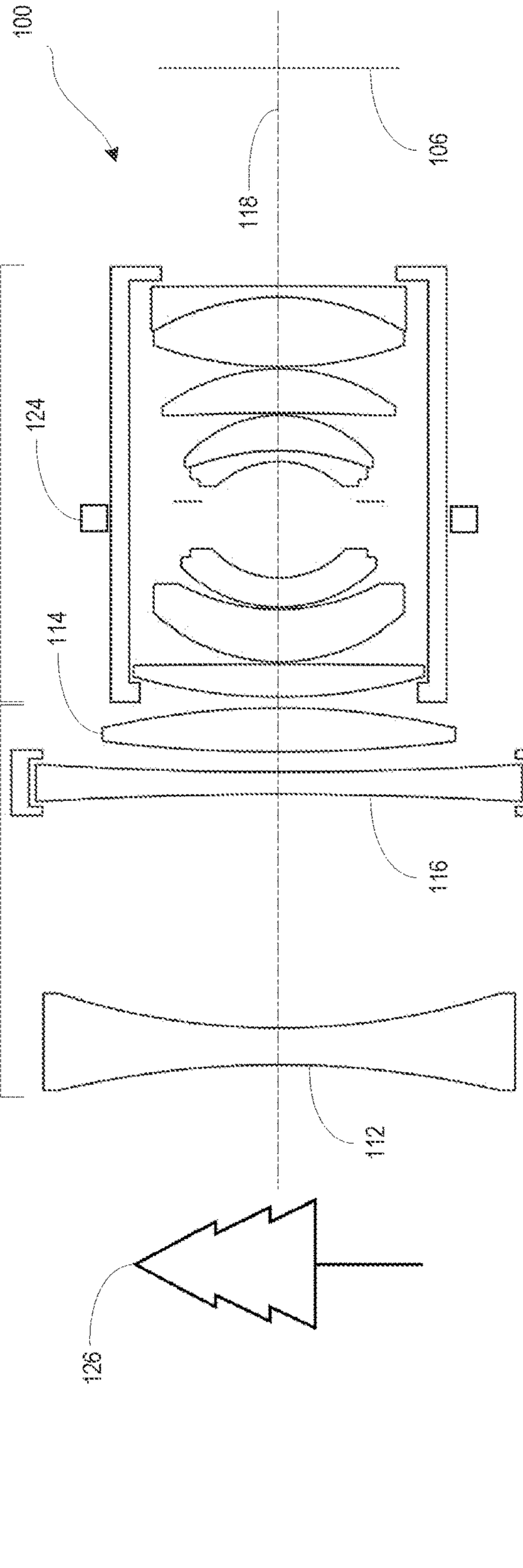


FIG. 2A
Infinity Focus

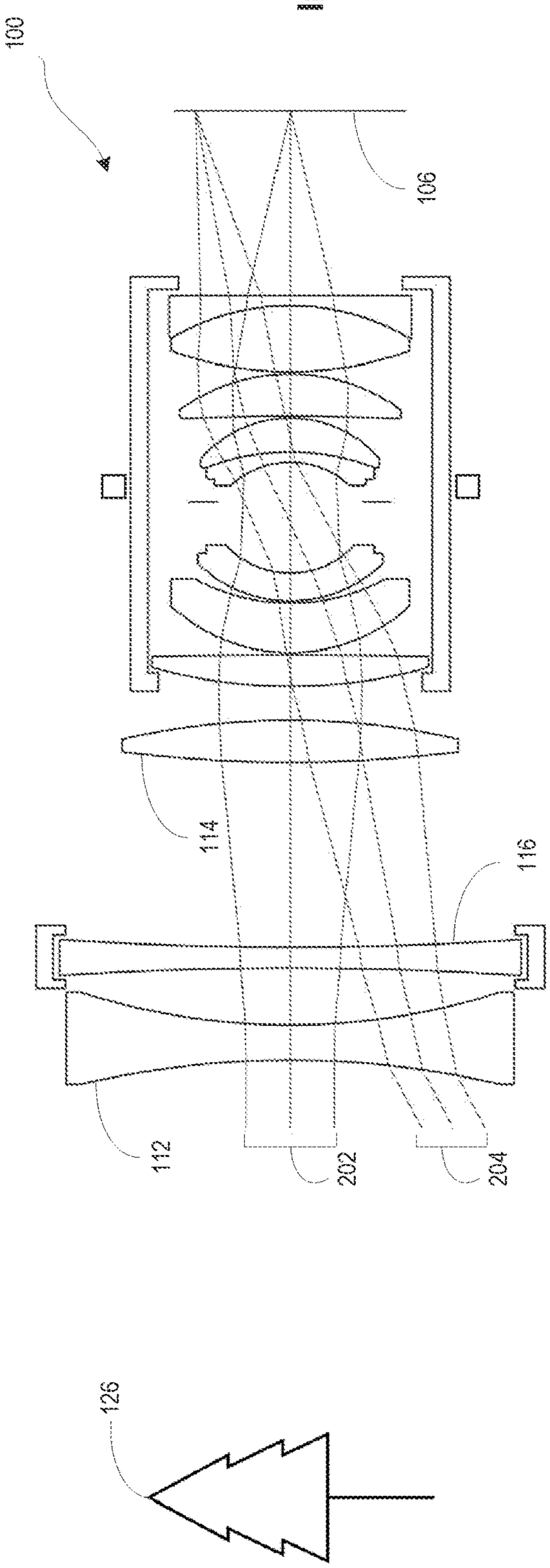
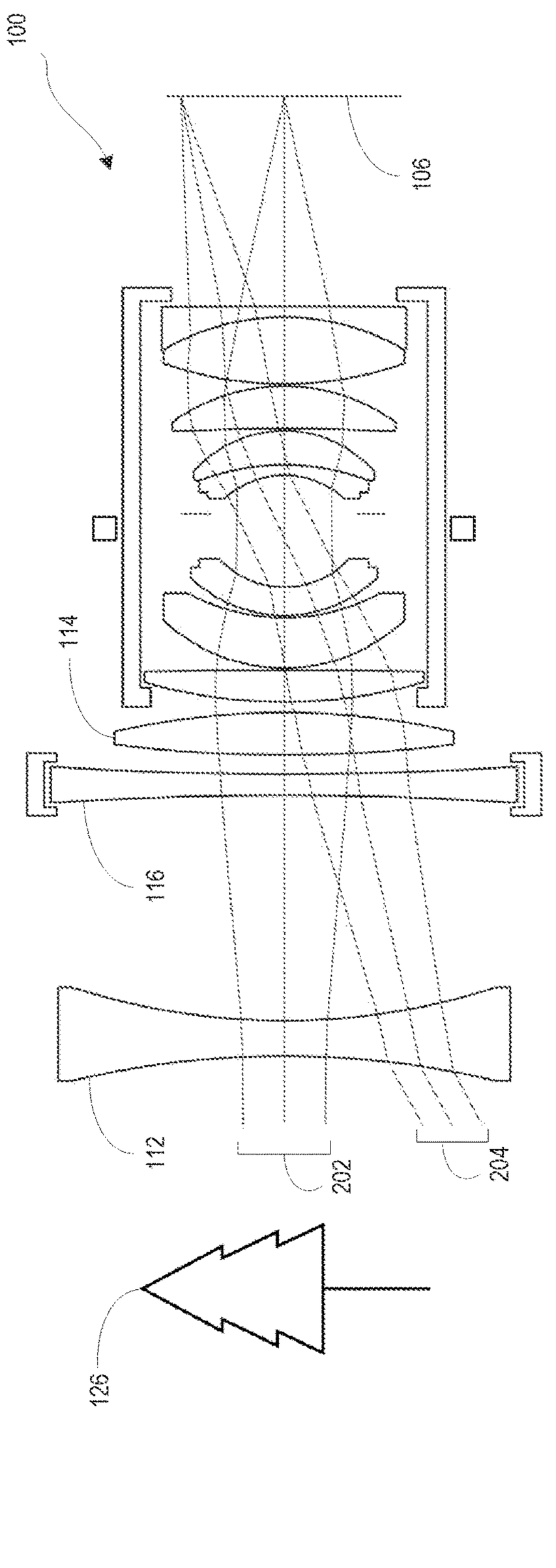


FIG. 2B
Close Focus



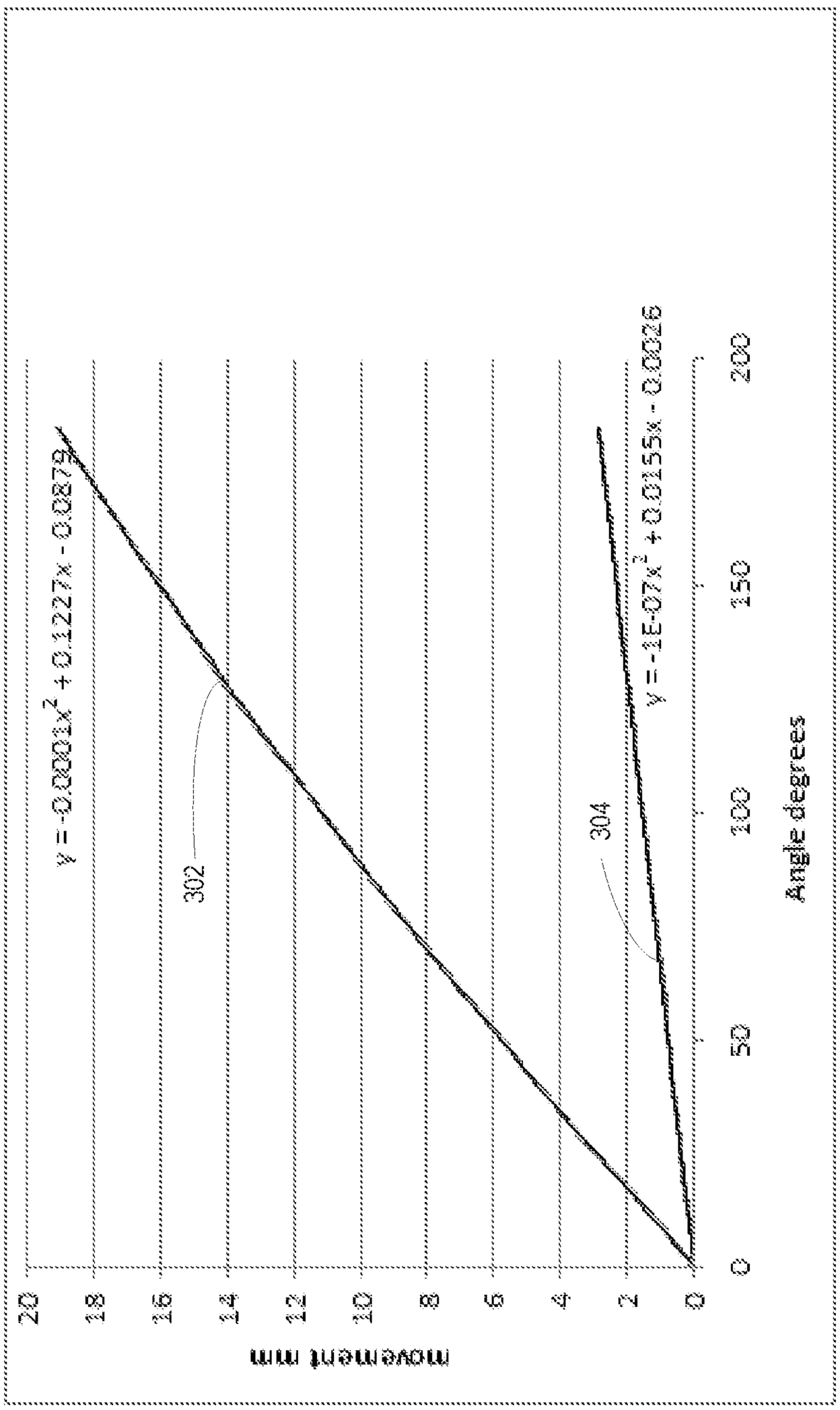

movement mm
0
2
4
6
8
10
12
14
16
18
20
302
304
y = -0.0801x² + 0.1227x - 0.0879
y = -1E-07x² + 0.0155x - 0.0026
50
100
150
200
Angle degrees

FIG. 3

FIG. 4A
Infinity Focus

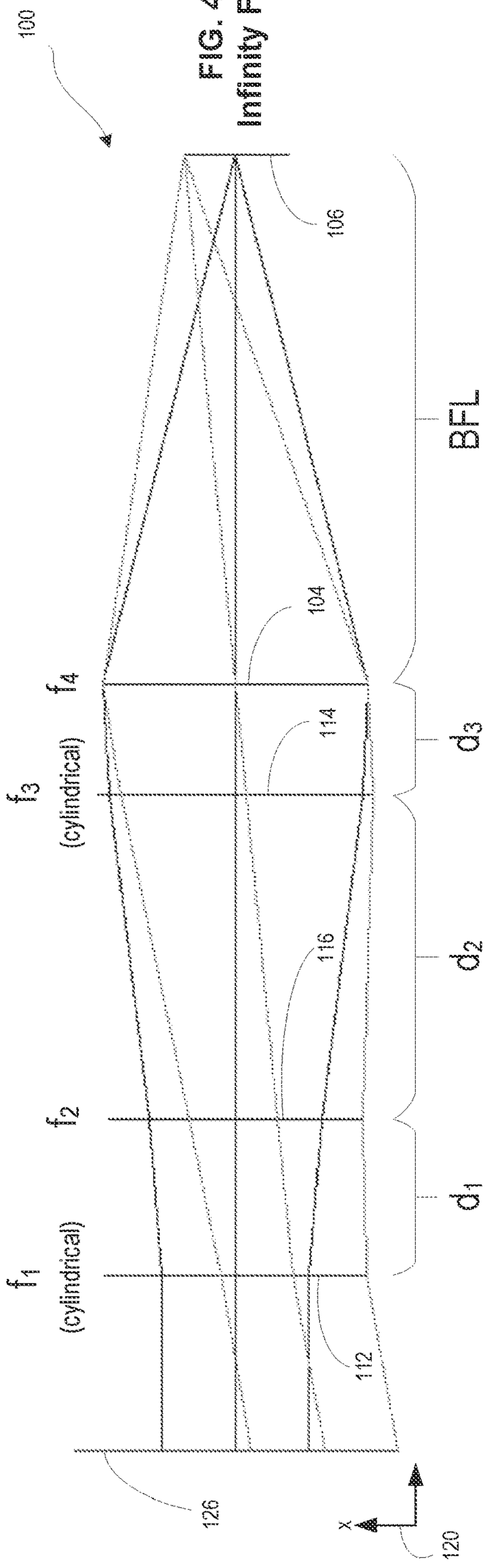


FIG. 4B
Close Focus

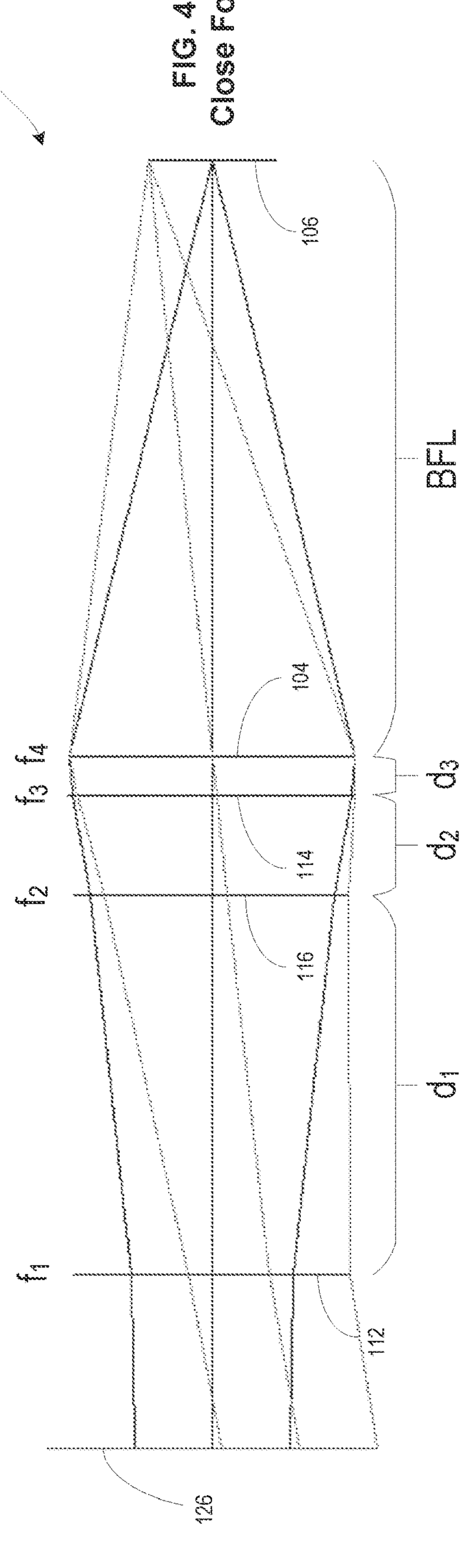


FIG. 4C
Infinity Focus

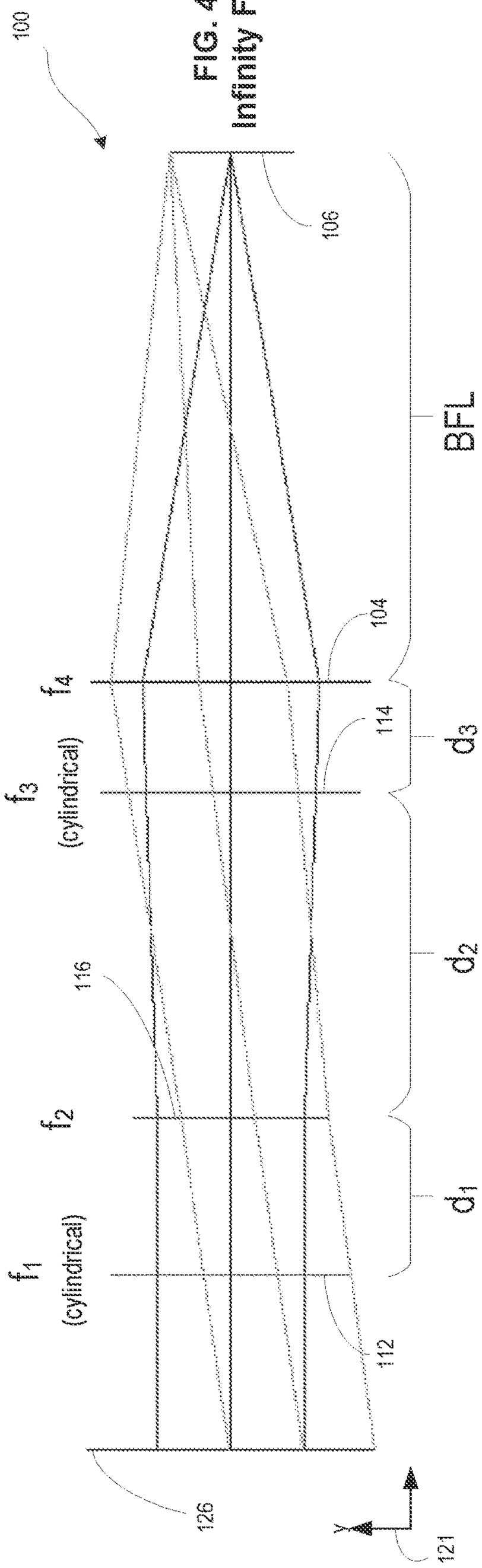


FIG. 4D
Close Focus

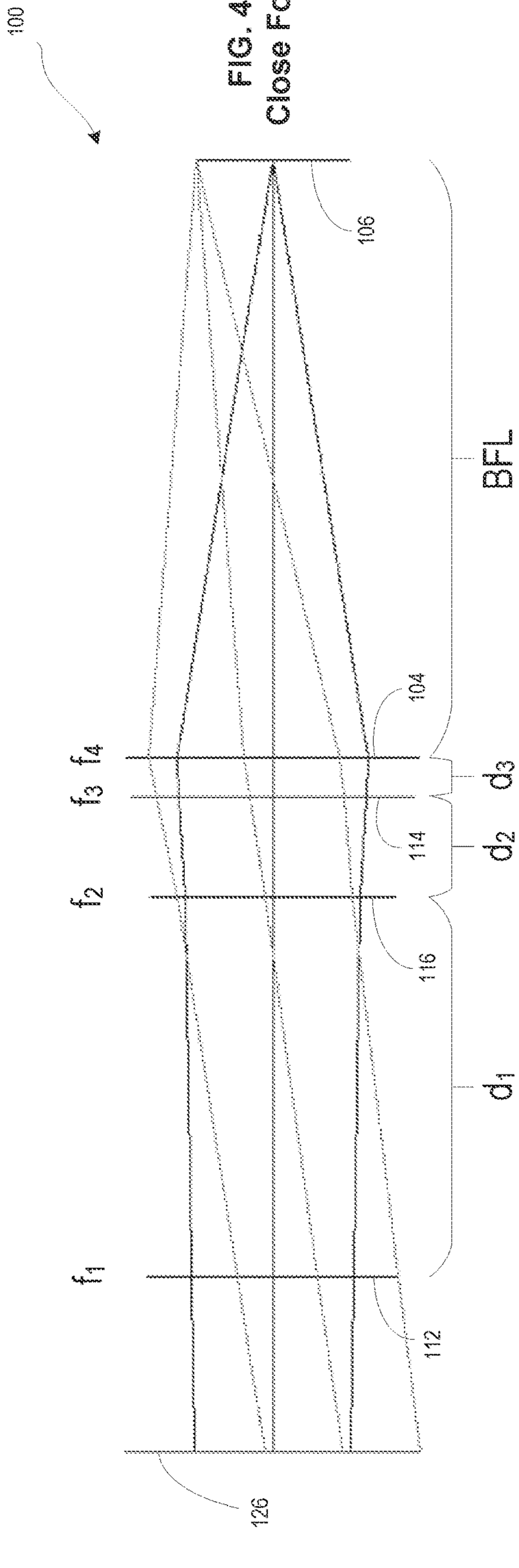


FIG. 5A
Infinity Focus

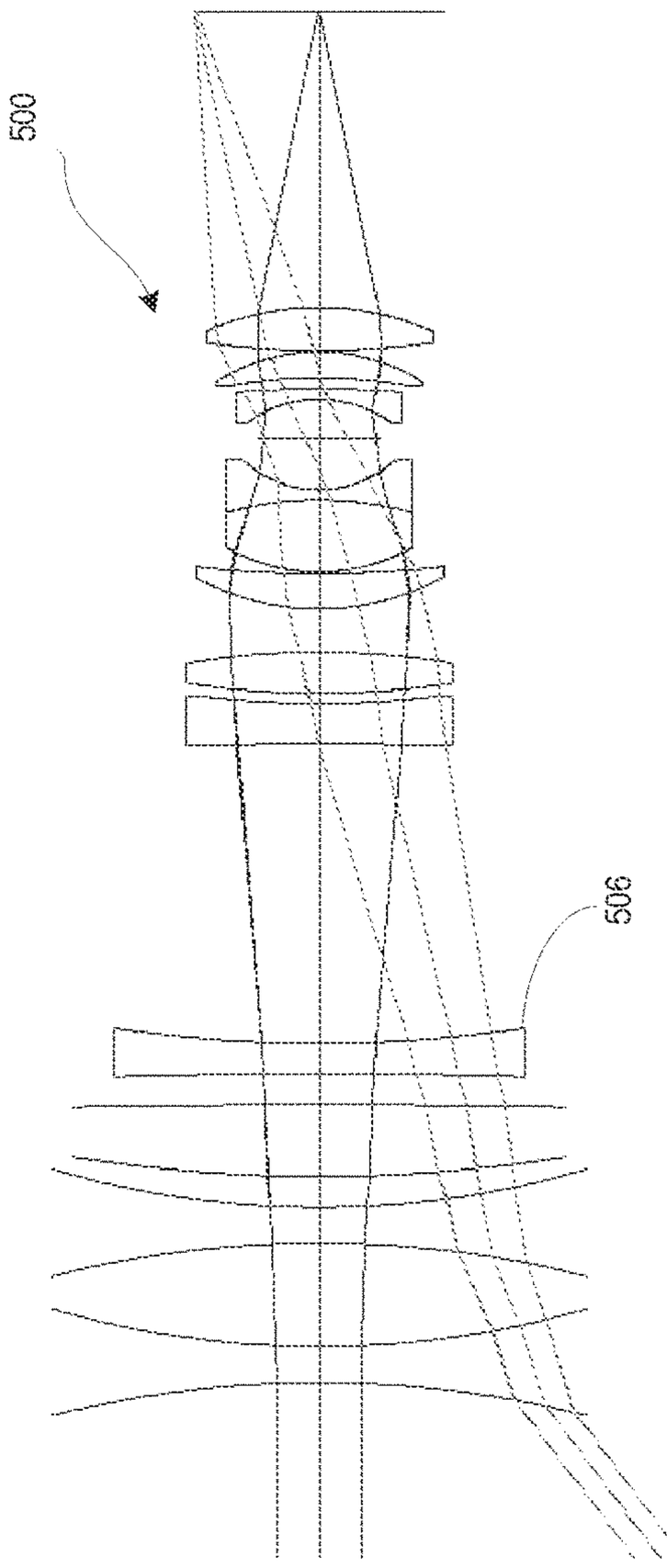


FIG. 5B
Close Focus

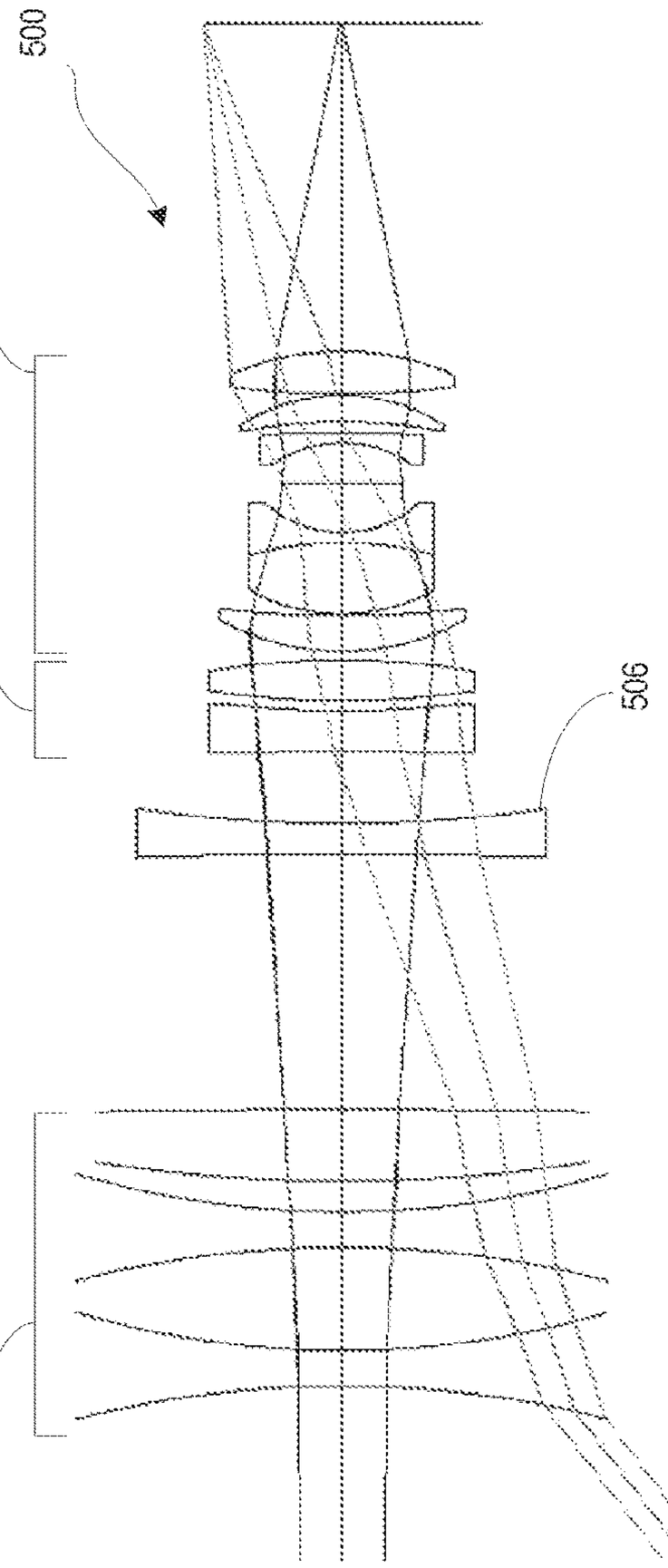


FIG. 6A
Infinity Focus

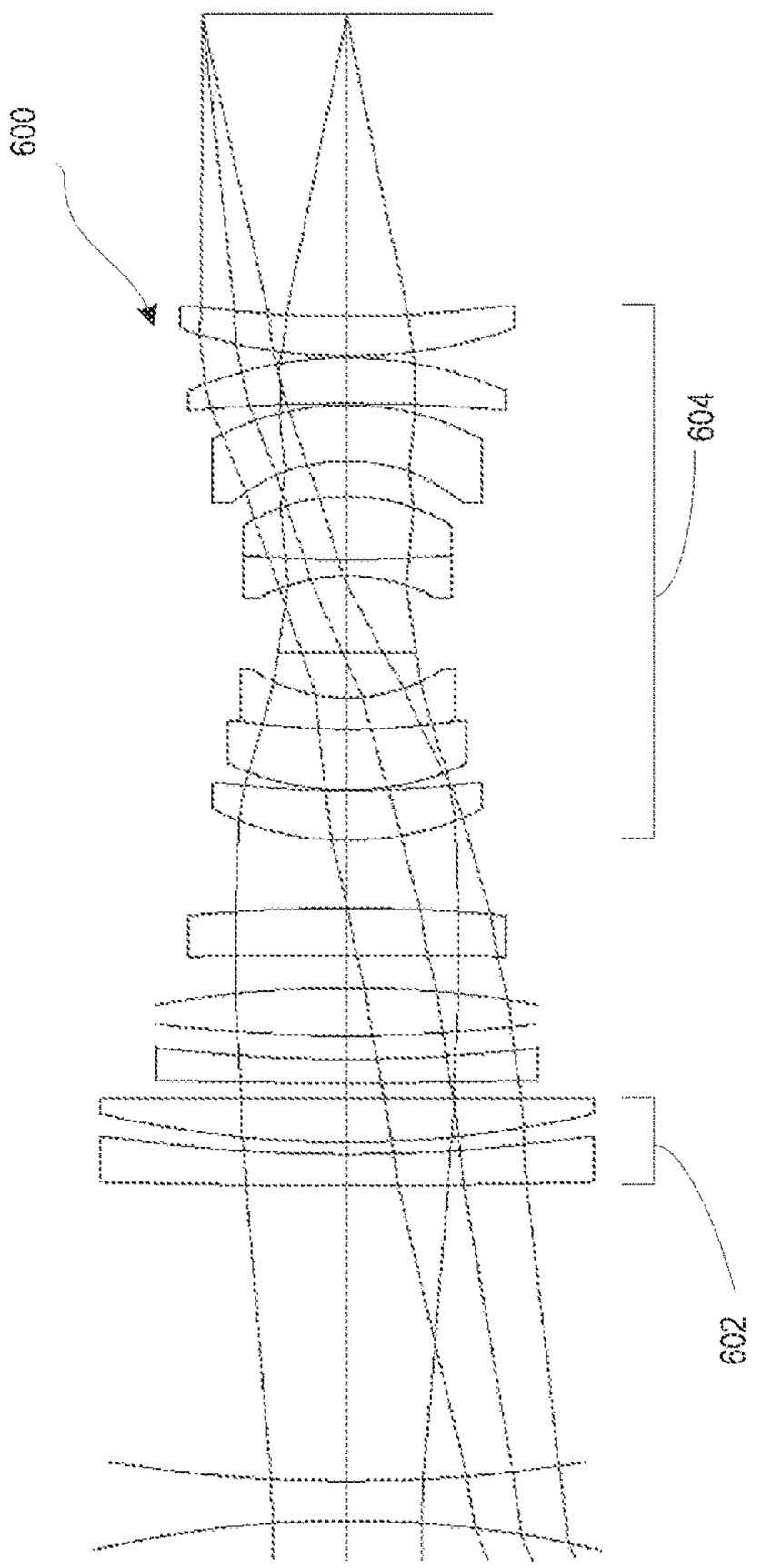


FIG. 6B
Close Focus

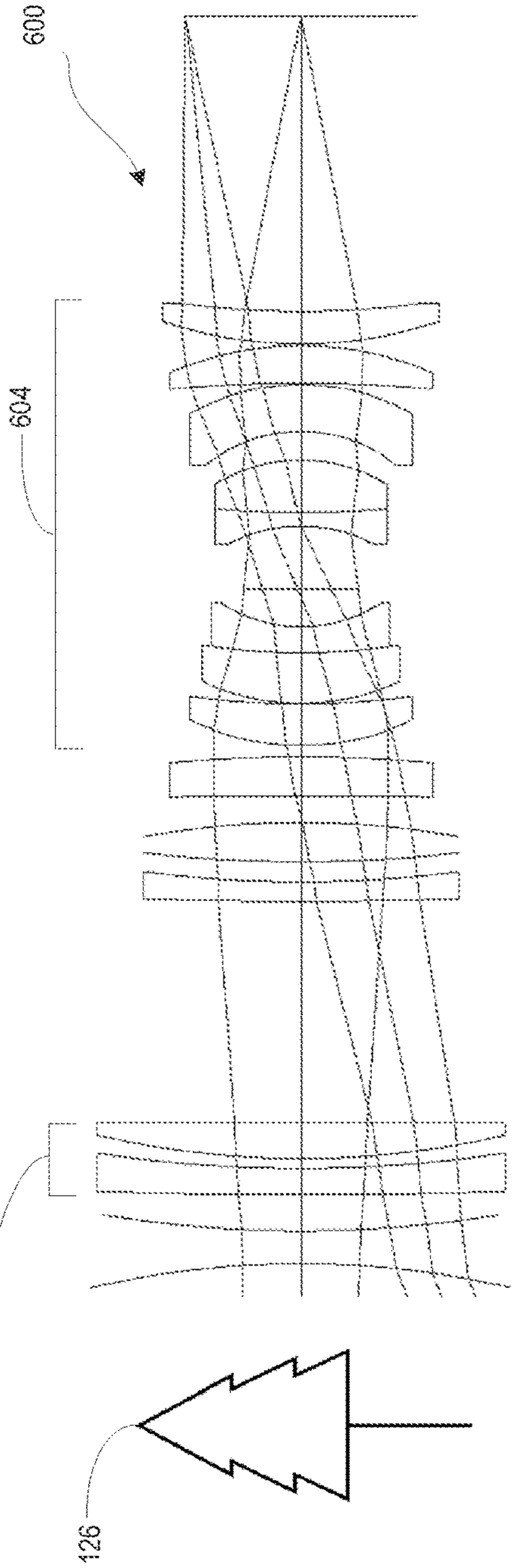


FIG. 7A
Infinity Focus

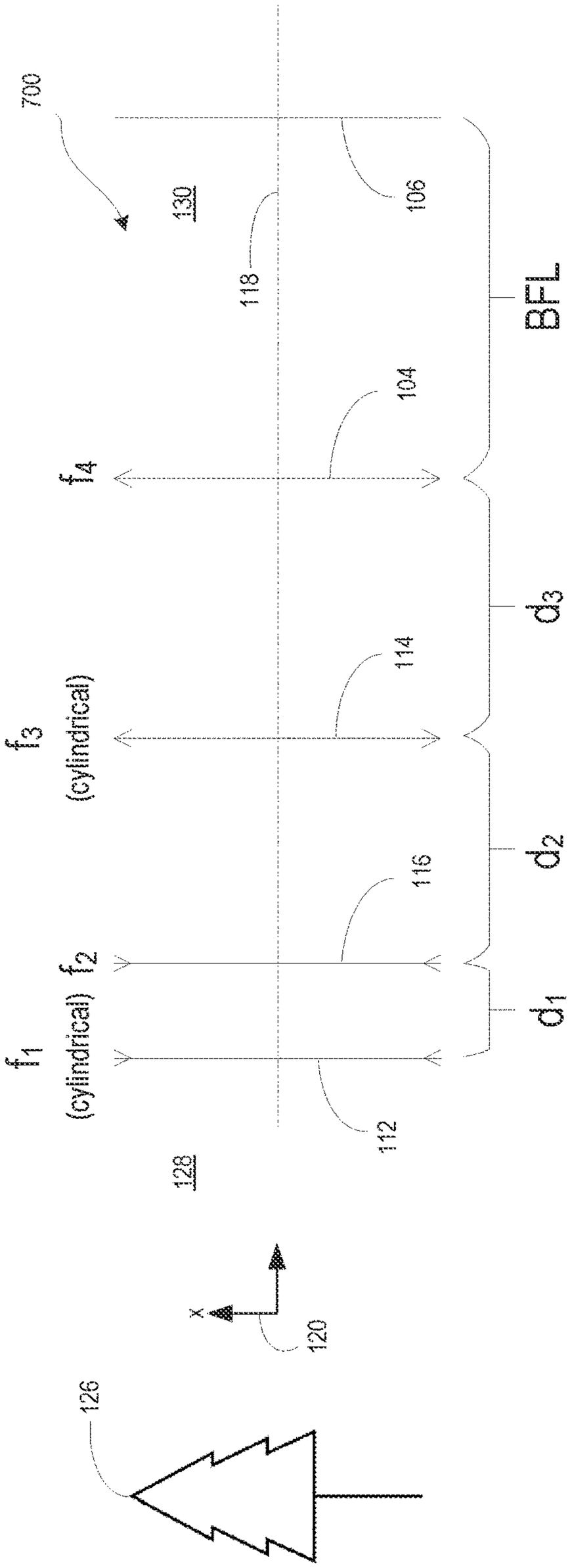


FIG. 7B
Close Focus

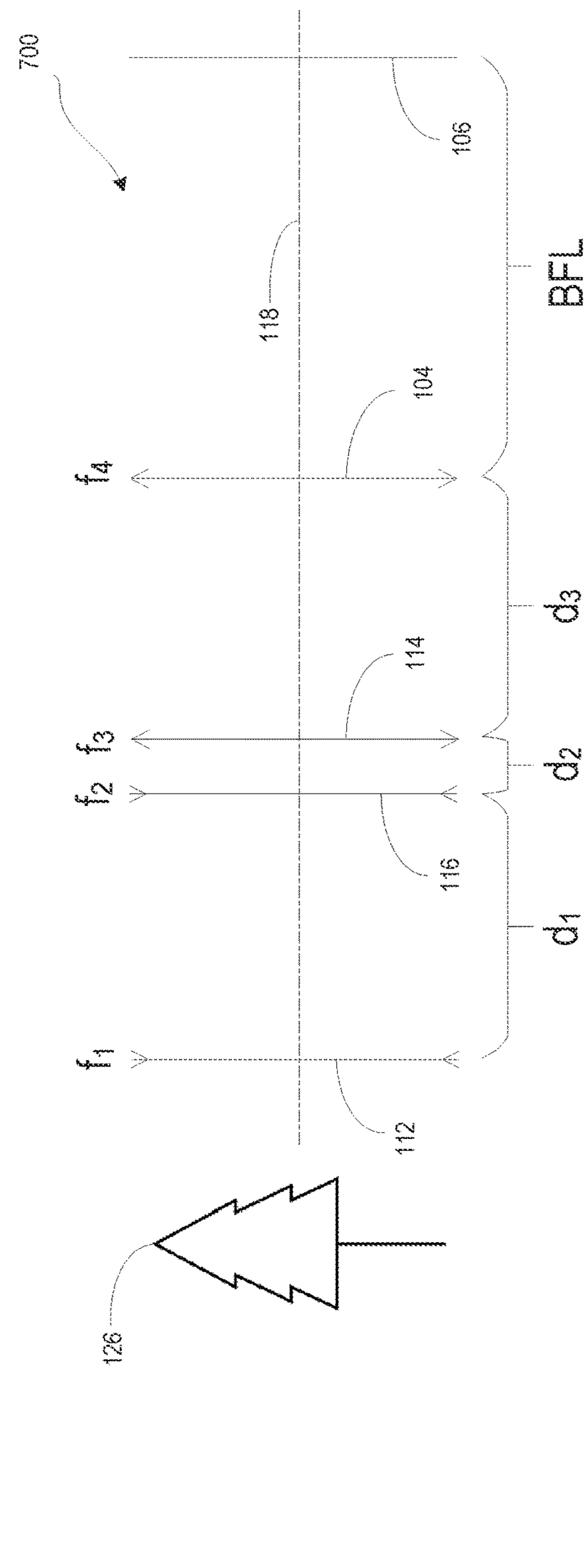


FIG. 8A
Infinity Focus

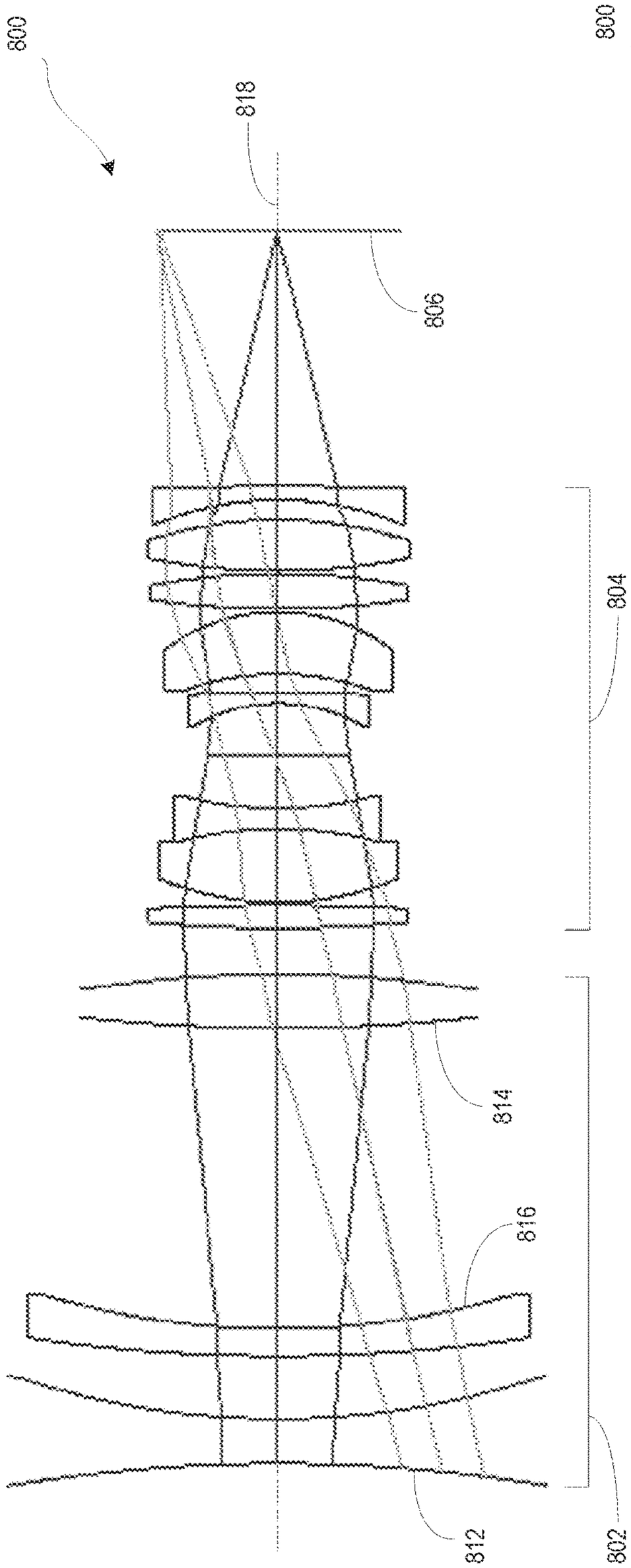


FIG. 8B
Close Focus

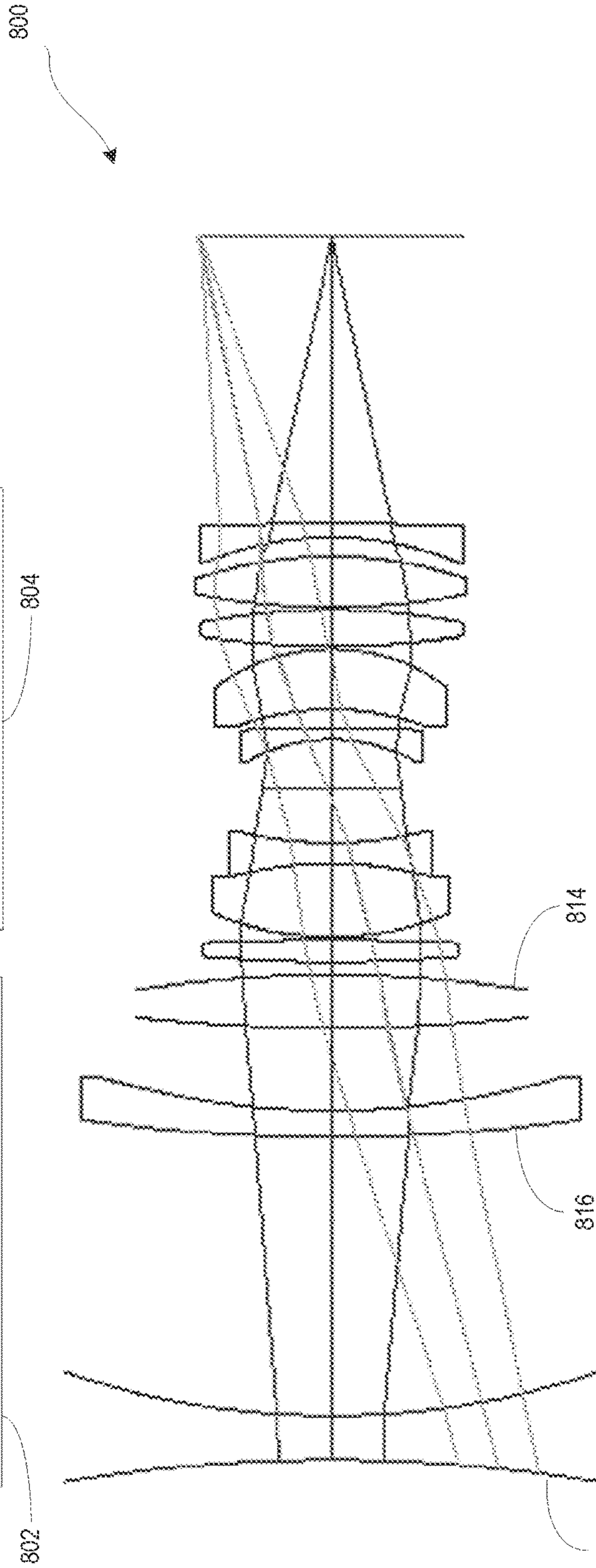


FIG. 9A
Infinity Focus

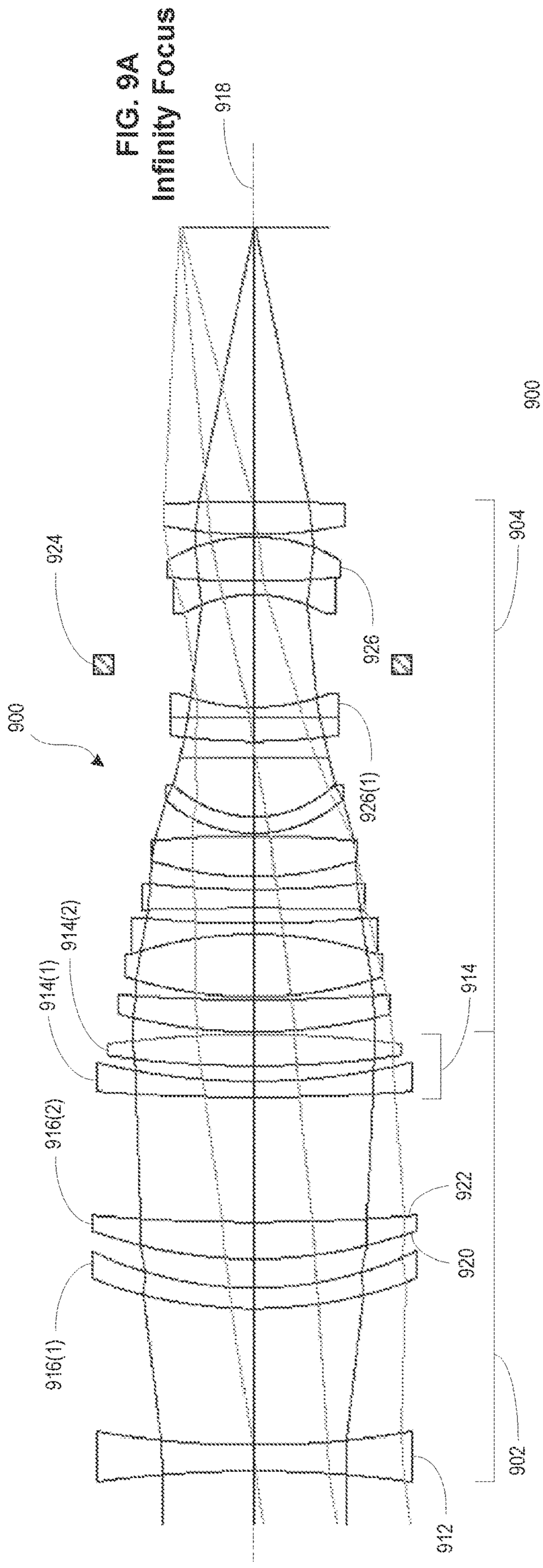
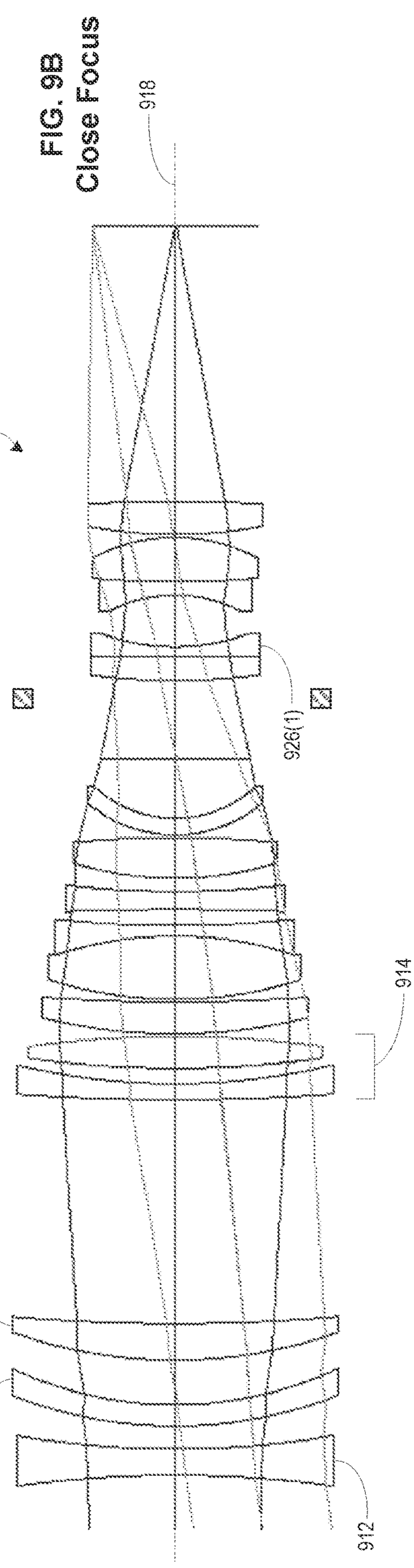


FIG. 9B
Close Focus



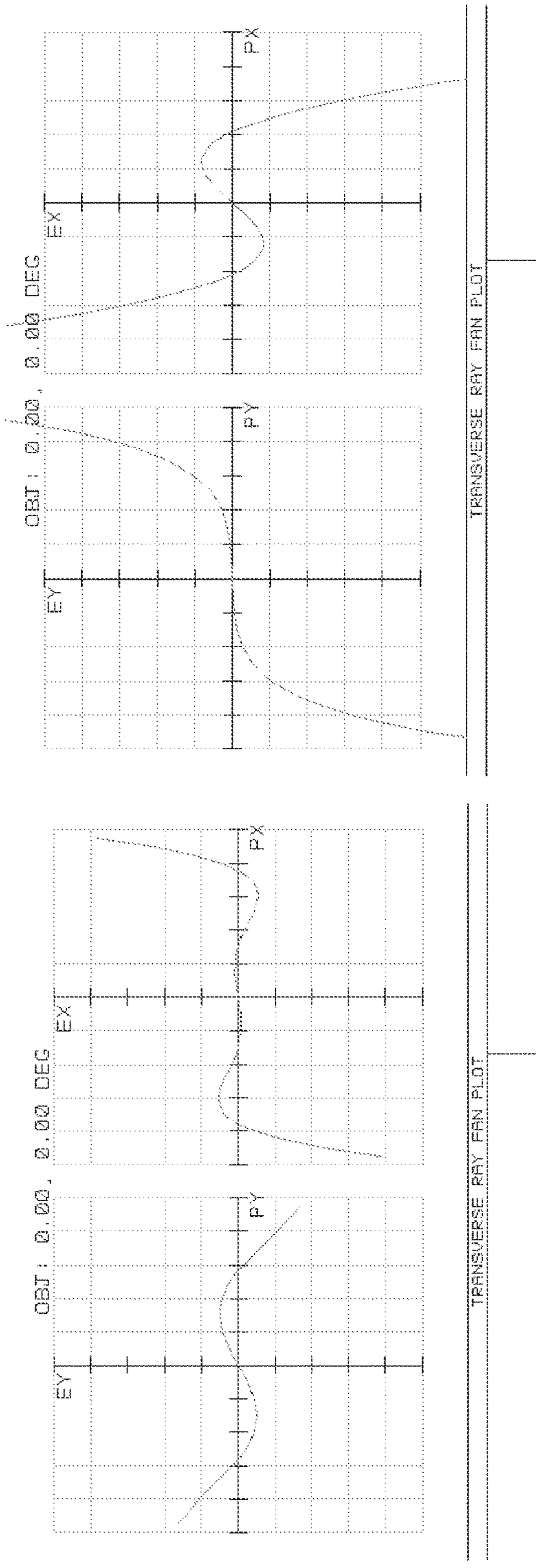


FIG. 10B
Close Focus

FIG. 10A
Infinity Focus

FIG. 11A
Infinity Focus

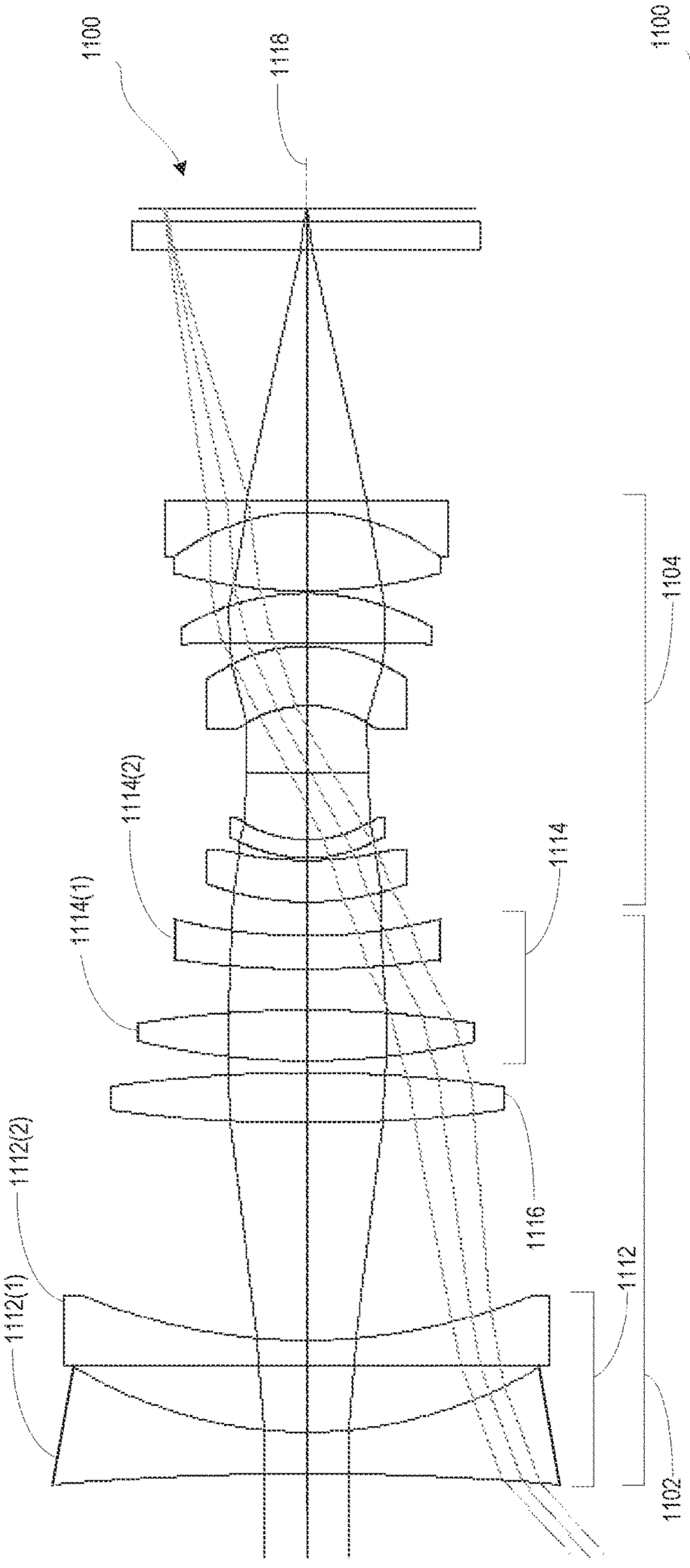
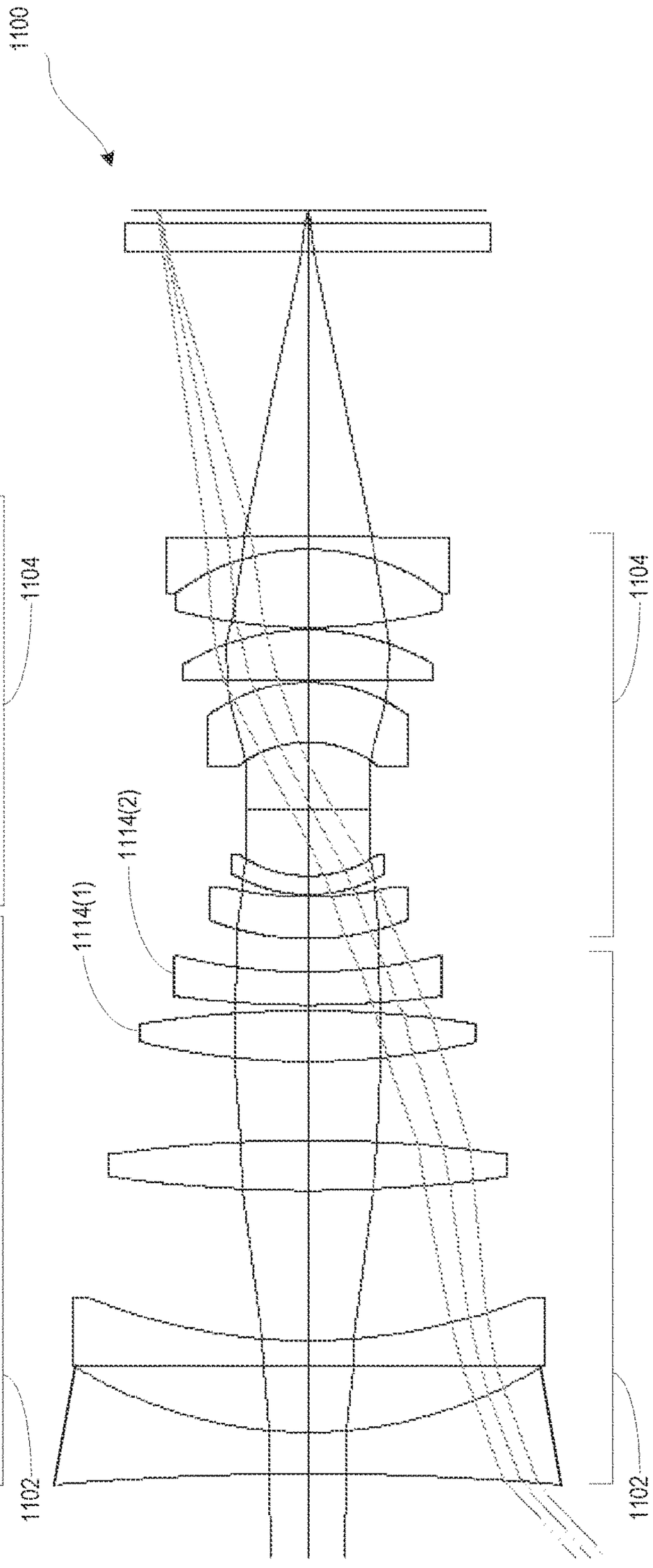


FIG. 11B
Close Focus



ANAMORPHIC LENS ASSEMBLIES

BACKGROUND

Anamorphic format is the cinematography technique of shooting a widescreen picture on standard 35 mm film or other visual recording media with a non-widescreen native aspect ratio. It also refers to the projection format in which a distorted image is stretched by an anamorphic projection lens to recreate the original aspect ratio on a viewing screen. An anamorphic lens typically includes a spherical primary lens, plus an anamorphic attachment (or an integrated lens element) that does the anamorphosing. The anamorphic element operates at infinite focal length, so that it has little or no effect on the focus of the primary lens it's mounted on, but still anamorphoses (distorts) the optical field. The distortion introduced in the camera must be corrected when the film is projected, so another lens is used in the projection booth that restores the picture back to its correct proportions to restore normal geometry. The picture is not manipulated in any way in the dimension that is perpendicular to the dimension that is anamorphosed.

Typically, an anamorphic lens captures (or projects) a wider horizontal angle of view than is normally possible with a spherical lens, in order to create a widescreen presentation. The anamorphic lens does this through optically distorting the image in the horizontal direction upon capture, and this distortion is then reversed in presentation. This method of widescreen image capture enables up to twice the width of the imager to be captured by distorting the image prior to recording, and then undistorting that compressed image later, either during post-production or during exhibition.

A traditional anamorphic lens optically compresses a wider angle of view onto a standard imager size by distorting the image's proportions, compressing the image horizontally. An alternative approach that achieves much the same result is to expand the image vertically. Either way, this horizontally squeezed (or vertically stretched) image is then undistorted into a widescreen aspect ratio through a corresponding anamorphic lens on a projector, or through digital correction of the distorted image.

An anamorphic lens assembly typically includes a spherical primary lens, plus an anamorphic attachment called an anamorphot (often an integrated multiple cylindrical-lens assembly) that does the squeezing (anamorphosing). The power of this attachment is typically zero in the vertical axis, such that it acts just like a piece of flat glass, and 0.5× in the horizontal axis, which reduces the effective focal length of the spherical lens by half in the horizontal direction. Most anamorphic systems work with this 0.5× compression (squeezing) power for gathering the image, which results in a 2× widening when presenting the image unsqueezed, although there are other compression ratios available, as well as the aforementioned vertical expansion approach. What this all means, generally, is that a 50 mm anamorphic lens will have the vertical angle of view of a 50 mm spherical lens, but the equivalent horizontal angle of view of a 25 mm spherical lens.

BRIEF DESCRIPTION OF DRAWINGS

Various embodiments in accordance with the present disclosure will be described with reference to the drawings, in which:

FIGS. **1**A and **1**B are schematic diagrams illustrating an anamorphic lens assembly according to some embodiments;

FIGS. **2**A and **2**B are schematic diagrams illustrating convergence of light rays through the anamorphic lens assembly of FIGS. **1**A and **1**B, respectively;

FIG. **3** is a graph illustrating two example polynomial relationships between a rotational angle of a focus ring and a position of a first spherical lens element in anamorphic lens assemblies according to some embodiments;

FIGS. **4**A and **4**B are schematic diagrams illustrating a general treatment of the anamorphic lens assembly of FIGS. **1**A and **1**B using paraxial optics, and viewed along the y-axis;

FIGS. **4**C and **4**D are schematic diagrams illustrating a general treatment of the anamorphic lens assembly of FIGS. **1**A and **1**B using paraxial optics, and viewed along the x-axis such that the cylindrical lenses have no optical power;

FIGS. **5**A and **5**B are schematic diagrams illustrating another anamorphic lens assembly according to some embodiments;

FIGS. **6**A and **6**B are schematic diagrams illustrating another anamorphic lens assembly according to some embodiments;

FIGS. **7**A and **7**B are schematic diagrams illustrating another anamorphic lens assembly according to some embodiments;

FIGS. **8**A and **8**B are schematic diagrams illustrating another anamorphic lens assembly according to some embodiments;

FIGS. **9**A and **9**B are schematic diagrams illustrating another anamorphic lens assembly according to some embodiments;

FIGS. **10**A and **10**B are ray aberration plots on-axis at infinity focus and close focus, respectively, for the anamorphic lens assembly of FIGS. **9**A and **9**B; and

FIGS. **11**A and **11**B are schematic diagrams illustrating another anamorphic lens assembly according to some embodiments.

DETAILED DESCRIPTION

The present disclosure relates to anamorphic lens assemblies. Traditionally, anamorphic lenses have different focal lengths along the horizontal and vertical axes because of the cylindrical lenses that perform the anamorphosing. These different focal lengths in perpendicular directions creates astigmatism. A lens with astigmatism is one in which light rays that propagate through the lens in two perpendicular planes have different foci (points where the light rays converge). For example, if a lens with astigmatism is used to form an image of a cross, the vertical and horizontal lines of the cross will be in sharp focus at two different distances.

Previous solutions to the astigmatism problem in anamorphic lenses have several drawbacks. For example, previous solutions have focused on adding cylindrical lenses to correct astigmatism, resulting in lens assemblies that are bulky, complicated, and expensive. These solutions have also created undesirable artifacts, such as the anamorphic mumps effect at close focus (e.g., at less than 10 feet from the object). For reasons of practical optics, the anamorphic squeeze is not uniform across the image field in traditional anamorphic systems (whether cylindrical, prismatic, or mirror-based). This variation results in some areas of the film image appearing more stretched than others. In the case of an actor's face, when positioned in the center of the screen, the face looks somewhat like the actor has the mumps, hence the name for the phenomenon.

Some of the present embodiments solve the above-described problems by providing an anamorphic lens assembly

having first and second cylindrical lens elements and a spherical lens element that is translatable along the optical axis of the anamorphic lens assembly between the first and second cylindrical lens elements. These embodiments use the movable spherical lens element to correct the astigmatism problem described above. These embodiments have a less complex structure as compared to previous anamorphic lens assemblies, which advantageously reduces the cost of producing anamorphic lens assemblies according to the present embodiments, as well as reducing their bulk. And, while previous solutions have tried to keep the anamorphic ratio constant as the distance between the camera and the object changed, some of the present embodiments allow the anamorphic ratio to change within a small range, such as less than 2% in some examples. This change of such small magnitude is imperceptible to the human eye, so that embodiments in which the change in the anamorphic ratio is below the threshold of 2% still advantageously produce no noticeable change to the anamorphic ratio as the distance between the camera and the object changes. In alternative embodiments, the anamorphic ratio may change within a somewhat larger range, such as less than 5%, or less than 4%, or less than 3%, in some examples. In some embodiments, the anamorphic ratio change with focus can be adjusted to match the characteristics of a given anamorphic lens as desired.

FIGS. **1**A and **1**B illustrate an anamorphic lens assembly **100** according to some embodiments. The anamorphic lens assembly **100** includes an anamorphic lens component **102** and a primary lens component **104**. In some embodiments, the anamorphic lens component **102** and the primary lens component **104** may be configured as a module that is attachable to, and removable from, other components of a camera system that includes an image plane **106**. In other embodiments, the anamorphic lens component **102** and the primary lens component **104** may be configured as separate modules that are attachable to, and removable from, one another, as well as attachable to, and removable from, other components of the camera system.

With reference to FIG. **1**A, the anamorphic lens component **102** may include a first housing **108** that forms the exterior of the anamorphic lens component **102**. Similarly, the primary lens component **104** may include a second housing **110** that forms the exterior of the primary lens component **104**. In some embodiments, the first and second housings **108**, **110** may be sections of an overall housing for the anamorphic lens assembly **100**. In the illustrated embodiment, only a portion of the first housing **108** is shown, and it should be understood that the first housing **108** at least partially surrounds and retains all lens elements of the anamorphic lens component **102**.

In FIG. **1**A the anamorphic lens assembly **100** is shown in an infinity focus arrangement, while in FIG. **1**B the anamorphic lens assembly **100** is shown in a close focus arrangement. With reference to FIG. **1**A, the anamorphic lens component **102** includes a first cylindrical lens element **112**, a second cylindrical lens element **114**, and a first spherical lens element **116** disposed between the first and second cylindrical lens elements **112**, **114**. Relative positions of the first and second cylindrical lens elements **112**, **114** are fixed along an optical axis **118** of the anamorphic lens assembly **100**, while the first spherical lens element **116** is translatable along the optical axis **118** with respect to the first and second cylindrical lens elements **112**, **114**. In some embodiments, the fixed distance between the first and second cylindrical lens elements **112**, **114**, coupled with the ability of the first spherical lens element **116** to translate along the optical axis **118** between the first and second cylindrical lens elements **112**, **114**, advantageously allows the spacings between the first spherical lens element **116** and the first and second cylindrical lens elements **112**, **114**, respectively, to be adjustable. This feature contributes to the advantageous optical characteristics of some of the present embodiments, as further described below.

In some embodiments, the first cylindrical lens element **112** has a first radius of curvature along a first axis, and the second cylindrical lens element **114** has a second radius of curvature along the first axis, where the first axis is perpendicular to the optical axis **118**. For example, in some embodiments the first and second cylindrical lens elements **112**, **114** have arcuate shapes including a concave portion or a convex portion on at least one side such that in the horizontal direction (e.g., along the x-axis **120**) one or both of the first and second cylindrical lens elements **112**, **114** increases or decreases the beam diameter by a refractive power provided by the arcuate shape(s) of the concave portion(s) or the convex portion(s), while in the vertical direction (e.g., along the y-axis) neither of the first and second cylindrical lens elements **112**, **114** has refractive power, or has negligible refractive power, or the first and second cylindrical lens elements **112**, **114** have equal and opposite refractive powers such that their combined refractive power is zero. In the illustrated embodiment, the first cylindrical lens element **112** has a negative power in the horizontal direction and the second cylindrical lens element **114** has a positive power in the horizontal direction. In alternative embodiments, however, the first and second cylindrical lens elements **112**, **114** may have any combination of powers, including negative, positive, and/or zero.

In some embodiments, the combined power of the first spherical lens element **116** and the first and second cylindrical lens elements **112**, **114** (the lenses of the anamorphic lens component **102**) is zero in the vertical direction and 0.5× in the horizontal direction. In other embodiments, however, the power of the anamorphic lens component **102** in the horizontal direction may be any other value, such as 0.75×, or 0.56×, or 0.33×, or 0.25×, or any other value. In still further embodiments, the power of the anamorphic lens component **102** is zero in the horizontal direction and 2× in the vertical direction (or 1.5×, or 1.8×, or 3×, or 4×, or any other value). The power of the anamorphic lens component **102** provides a desired amount of squeezing or stretching (anamorphosing) along a desired axis to achieve the specified anamorphic format.

In the embodiment illustrated in FIGS. **1**A and **1**B, each of the first spherical lens element **116**, the first cylindrical lens element **112**, and the second cylindrical lens element **114** is represented as a single lens (singlet). This embodiment is, however, merely one example. In alternative embodiments, any of the lens elements **112**, **114**, **116** may comprise multiple lenses (e.g., a lens group), such as pairs of lenses (doublets). For example, in some embodiments, the first spherical lens element **116** may comprise a singlet while the first and second cylindrical lens elements **112**, **114** may comprise doublets. In another example, in some embodiments, the first spherical lens element **116** may comprise a doublet while the first and second cylindrical lens elements **112**, **114** may comprise singlets.

With reference to FIG. **1**A, the primary lens component **104** is disposed on a side of the second cylindrical lens element **114** opposite the first spherical lens element **116**, and includes one or more second spherical lens elements **122**. The second spherical lens elements **122** may be configured to provide primary imaging by, for example, varying

a focus of an image formed by the primary lens component **104**. The one or more second spherical lens elements **122** may therefore be translatable along the optical axis **118** with respect to the first and second cylindrical lens elements **112**, **114**, as further described below. In some embodiments, the second spherical lens elements **122** may be configured to vary other optical properties of the image, such as a softness of the image, a size of the image, or may be configured to correct for blur or aberrations in the image, or to change other optical properties as desired. While the second spherical lens elements **122** are described herein as being spherical, in some embodiments one or more of the second spherical lens elements **122** may be aspherical if desired.

With reference to FIG. **1**A, and as described above, the relative positions of the first and second cylindrical lens elements **112**, **114** are fixed along the optical axis **118** of the anamorphic lens component **102**. The first spherical lens element **116** is, however, translatable along the optical axis **118** with respect to the first and second cylindrical lens elements **112**, **114**. For example, FIG. **1**A illustrates the anamorphic lens assembly **100** in an infinity focus arrangement in which the first spherical lens element **116** is disposed at the limit of its travel in the direction of the first cylindrical lens element **112**, while FIG. **1**B illustrates the anamorphic lens assembly **100** in a close focus arrangement in which the first spherical lens element **116** is disposed at the limit of its travel in the direction of the second cylindrical lens element **114**.

In some embodiments, movement of the first spherical lens element **116** may be controlled by a focus adjustment member **124** (e.g., a focus ring) disposed around the primary lens component **104**. In embodiments in which the focus adjustment member **124** is a focus ring, the focus ring **124** may be rotatable about the primary lens component **104**. The focus ring **124** may be mechanically coupled to one or more additional focus adjustment members (not shown), which may be mechanically coupled to the first spherical lens element **116**, such that rotation of the focus ring **124** adjusts the focus of the primary lens component **104** while simultaneously inducing translation of the first spherical lens element **116** along the optical axis **118** between the first and second cylindrical lens elements **112**, **114**. Together, the focus adjustment member **124** and the one or more additional focus adjustment members that mechanically couple the focus adjustment member **124** to the first spherical lens element **116** may comprise a translation mechanism of the first spherical lens element **116**. Thus, for example, as the focus ring **124** is rotated in a first rotational direction around the primary lens component **104**, the primary lens component **104** may be adjusted away from the infinity focus arrangement (FIG. **1**A) and toward the close focus arrangement (FIG. **1**B) while the first spherical lens element **116** travels away from the first cylindrical lens element **112** and toward the second cylindrical lens element **114**. Conversely, as the focus ring **124** is rotated in a second rotational direction around the primary lens component **104**, the primary lens component **104** may be adjusted away from the close focus arrangement (FIG. **1**B) and toward the infinity focus arrangement (FIG. **1**A) while the first spherical lens element **116** travels away from the second cylindrical lens element **114** and toward the first cylindrical lens element **112**. In some embodiments, focus of the primary lens component **104** may be controlled by a threaded focus adjustment mechanism, or controlled by a cam mechanism. For example, as described below, in some embodiments relative movements of the first spherical lens element **116** and the primary lens component **104** may be nonlinear. In such embodiments, one of the first spherical lens element **116** or the primary lens component **104** may move linearly with a thread, while the other may move via a cam mechanism.

With reference to FIG. **1**A, the anamorphic lens assembly **100** according to some embodiments is configured to produce a focused image of an object **126** (on an object side **128** of the anamorphic lens assembly **100**) at the image plane **106** of a camera (on an image side **130** of the anamorphic lens assembly **100**). In some embodiments, the camera may be a digital camera, and the image plane **106** may comprise an image sensor having an imaging area for receiving images from the anamorphic lens assembly **100**. In other embodiments, the camera may be a film camera, and the image plane **106** may comprise film having an imaging area for receiving, on film, images from the anamorphic lens assembly **100**. In some embodiments, the anamorphic lens assembly **100** may be configured as a module that is attachable to, and removable from, the camera, while in other embodiments the anamorphic lens assembly **100** may be integrated within the camera. In some embodiments, the camera may be a cinema camera configured to generate images for presentation by a cinema projector. In some embodiments, the primary lens component **104** may have an internal focus mechanism (not shown) that enables one or more lenses within the primary lens component **104** to move in order to achieve focus.

In some embodiments, as the first spherical lens element **116** travels along the optical axis **118** relative to the first and second cylindrical lens elements **112**, **114**, a spacing between the second cylindrical lens element **114** and the primary lens component **104** changes. For example, with reference to both FIGS. **1**A and **1**B, as the first spherical lens element **116** travels away from the infinity focus arrangement of FIG. **1**A (away from the first cylindrical lens element **112**) and toward the close focus arrangement of FIG. **1**B (toward the second cylindrical lens element **114**), a spacing between the second cylindrical lens element **114** and the primary lens component **104** decreases. This relative movement of the first spherical lens element **116** and the primary lens component **104** maintains focus of the image at the image plane **106**, and in some embodiments the image plane **106** may move relative to the primary lens component **104** as the anamorphic lens assembly **100** transitions between the infinity focus arrangement and the close focus arrangement. In alternative embodiments, such as where the first spherical lens element **116** has positive refractive power, the relative directions of movement of the second cylindrical lens element **114** and the primary lens component **104** may be reversed, such that as the first spherical lens element **116** travels away from the infinity focus arrangement and toward the close focus arrangement, a spacing between the second cylindrical lens element **114** and the primary lens component **104** increases.

In some embodiments, relative movements of the first spherical lens element **116** and the focus ring **124**, and relative movements of the primary lens component **104** and the focus ring **124**, may be defined by polynomial relationships. For example, with reference to FIG. **3**, two example polynomial relationships are plotted with the x-axis representing the rotational angle of the focus ring **124** and the y-axis representing the positions of the first spherical lens element **116** and the primary lens component **104** along the optical axis **118**. In some embodiments, the positions of the first spherical lens element **116** and the primary lens component **104** along the optical axis **118** may be measured with reference to respective origin points at the respective limits

of travel for each of the first spherical lens element **116** and the primary lens component **104** corresponding to the infinity focus arrangement of FIG. **1**A. Thus, as the focus ring **124** is rotated in the first rotational direction around the primary lens component **104**, the rotational angle of the focus ring **124** increases (positive movement along the x-axis) while the first spherical lens element **116** and the primary lens component **104** both move along the optical axis **118** away from the infinity focus arrangement (FIG. **1**A, origin point) and toward the close focus arrangement (FIG. **1**B, terminal point). A first one **302** of the example polynomial relationships is $y=-0.0001x^2+0.1227x-0.0879$, and represents the relative movements of the focus ring **124** and the first spherical lens element **116**, while a second one **304** of the example polynomial relationships is $y=-0.0000001x^2+0.0155x-0.0026$, and represents the relative movements of the focus ring **124** and the primary lens component **104**. It will be appreciated, however, that these equations are only examples and are in no way limiting. In some embodiments, the movements of the first spherical lens element **116** and the primary lens component **104** are related such that the combination of all of the optical elements maintains a constant (or nearly constant) anamorphic ratio and focus along the x-axis and the y-axis.

With reference to both FIGS. **1**A and **1**B, as the object **126** moves from a first position (FIG. **1**A) toward the anamorphic lens assembly **100**, astigmatism is created by the primary lens component **104** due to different focal lengths along the horizontal and vertical axes. Therefore, refocusing only the primary lens component **104** will not achieve good focus at the image plane **106**. In some embodiments, the ability of the first spherical lens element **116** to translate along the optical axis **118** relative to the cylindrical lens elements **112**, **114** and the primary lens component **104** creates an opposite astigmatism to that created by the movement of the object **126** relative to the primary lens component **104**, enabling the anamorphic lens assembly **100** to achieve good focus at the image plane **106**, regardless of the distance of the object **126** to the lens assembly **100**.

In some embodiments, optical characteristics of the first cylindrical lens element **112**, the first spherical lens element **116**, and the second cylindrical lens element **114** in combination produce zero astigmatism at the image plane **106** for the object **126** at infinity focus. For example, as shown in FIG. **2**A, when the object **126** is at infinity focus and the anamorphic lens assembly **100** is in the infinity focus arrangement, on-axis light rays **202** converge at the image plane **106** while off-axis light rays **204** similarly converge at the image plane **106**. In some embodiments, the optical characteristics of the first cylindrical lens element **112**, the first spherical lens element **116**, and the second cylindrical lens element **114** in combination are adjustable as the first spherical lens element **116** translates along the optical axis **118** with respect to the first and second cylindrical lens elements **112**, **114**, such that the first cylindrical lens element **112**, the first spherical lens element **116**, and the second cylindrical lens element **114** in combination produce a first astigmatism that is opposite to a second astigmatism produced by unequal movement of the image plane **106** along horizontal and vertical axes as the object **126** moves from a first position at the infinity focus toward the anamorphic lens assembly **100**. For example, as shown in FIG. **2**B, when the object **126** is at close range and the anamorphic lens assembly **100** is in the close focus arrangement, on-axis light rays **202** converge at the image plane **106** while off-axis light rays **204** similarly converge at the image plane **106**.

In some embodiments, a spherical aberration of the first spherical lens element **116** is corrected to match optical characteristics of the entire anamorphic lens assembly **100**. In some embodiments, because the first spherical lens element **116** is movable with respect to the first and second cylindrical lens elements **112**, **114** and the primary lens component **104**, its spherical aberration cannot be perfectly corrected for every position of the first spherical lens element **116** along the optical axis **118**. Thus, there may be no ideal shape for the first spherical lens element **116**. Rather, its shape is selected to balance other aberrations from the first and second cylindrical lens elements **112**, **114** and the primary lens component **104** based on the optical characteristics of those lenses.

With reference to both FIGS. **1**A and **1**B, as the anamorphic lens assembly **100** transitions between the infinity focus arrangement (FIG. **1**A) and the close focus arrangement (FIG. **1**B), the horizontal and vertical focal lengths of the combination of the first cylindrical lens element **112**, the first spherical lens element **116**, and the second cylindrical lens element **114** change due to the changes in the spacings between these lens elements. The changing focal lengths cause the anamorphic ratio of the anamorphic lens component **102** to also change slightly. In some embodiments the difference in the anamorphic ratio is less than 2% (or less than 5%, or less than 4%, or less than 3%) between the infinity focus arrangement and the close focus arrangement.

In the illustrated embodiment, the first spherical lens element **116** has negative refractive power, and the first spherical lens element **116** moves away from the first cylindrical lens element **112** and toward the second cylindrical lens element **114** as the anamorphic lens assembly **100** transitions away from the infinity focus arrangement and toward the close focus arrangement. In alternative embodiments, the first spherical lens element **116** may have positive refractive power, and in such embodiments the first spherical lens element **116** would move away from the second cylindrical lens element **114** and toward the second first cylindrical lens element **112** as the anamorphic lens assembly **100** transitions away from the infinity focus arrangement toward the close focus arrangement.

FIGS. **4**A-**4**D illustrate a general treatment of the anamorphic lens assembly **100** of FIGS. **1**A and **1**B using paraxial optics. In FIGS. **4**A and **4**B, the viewpoint is along the y-axis **121**, while in FIGS. **4**C and **4**D the viewpoint is along the x-axis **120**. Each of the lens elements/components **104**, **112**, **114**, **116** is illustrated as a thin lens with a focal length f_x. The distances between the lens elements/components **104**, **112**, **114**, **116** are represented as d_x, where d_1 is the distance between the first cylindrical lens element **112** and the first spherical lens element **116**, d_2 is the distance between the first spherical lens element **116** and the second cylindrical lens element **114**, and d_3 is the distance between the second cylindrical lens element **114** and the primary lens component **104**. The back focal length (BFL) of the anamorphic lens assembly **100** is the distance between the primary lens component **104** and the image plane **106**.

In the illustrated embodiment, the first cylindrical lens element **112** has a negative focal length f_1 along the x-axis **120**, the first spherical lens element **116** has a negative focal length f_2, the second cylindrical lens element **114** has a positive focal length f_3 along the x-axis **120**, and the primary lens component **104** has a positive focal length f_4. In some embodiments, the focal lengths of the lens elements/components **104**, **112**, **114**, **116** are selected such that the effective focal length of the anamorphic lens assembly **100** along the x-axis **120** (horizontal) is shorter than the effective

focal length along the y-axis (vertical) by the desired anamorphic ratio. In some embodiments, the BFL of the anamorphic lens assembly **100** is identical along both the x-axis **120** and the y-axis (zero astigmatism).

The EFL (effective focal length) of the anamorphic lens assembly **100** along the y-axis can be calculated from the focal lengths of the first spherical lens element **116** (f_2) and the primary lens component **104** (f_4) and the distance between them d_2+d_3. For the y-axis case, the first and second cylindrical lens elements **112**, **114** can be ignored, since they have zero power along the y-axis (at least in this example embodiment). The formula for the EFL of the first spherical lens element **116** and the primary lens component **104** along the y-axis (when represented as two thin lenses) is:

$$EFL_y = \frac{1}{f_2} + \frac{1}{f_4} - \frac{d_2 + d_3}{f_2 \times f_4},$$

where d_2+d_3 represents the total distance between the first spherical lens element **116** and the primary lens component **104**.

Calculation of the EFL of the anamorphic lens assembly **100** along the x-axis **120** (which includes the cylindrical powers of the first and second cylindrical lens elements **112**, **114**) is more complex, but can be represented using ABCD matrix techniques. For the anamorphic lens assembly **100**, which includes four lens elements/components **104**, **112**, **114**, **116**, the Gaussian transfer matrix is represented by the following product:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{-1}{f_4} & 1 \end{pmatrix} \times \begin{pmatrix} 1 & d_3 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ \frac{-1}{f_3} & 1 \end{pmatrix} \times \begin{pmatrix} 1 & d_2 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ \frac{-1}{f_2} & 1 \end{pmatrix} \times \begin{pmatrix} 1 & d_1 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ \frac{-1}{f_1} & 1 \end{pmatrix}$$

The resultant ABCD transfer matrix can be used to determine the EFL of the anamorphic lens assembly **100** along the x-axis **120**, which is represented by the C term (lower left) of the matrix. The formula for the EFL of the anamorphic lens assembly **100** along the x-axis **120** is.

$$\frac{-1}{EFL_x} = \frac{-1}{f_4} - \frac{1}{f_3}\left(1 - \frac{d_3}{f_3}\right) - \frac{1}{f_2}\left(d_2\left(\frac{-1}{f_3}\left(1 - \frac{d_3}{f_4}\right) - \frac{1}{f_4}\right) + 1 - \frac{d_3}{f_4}\right) - \frac{1}{f_1}\left(d_1\left(\frac{-1}{f_3}\left(1 - \frac{d_3}{f_4}\right) - \frac{1}{f_4} - \frac{1}{f_2}\left(d_2\left(\frac{-1}{f_4} - \frac{1}{f_3}\left(1 - \frac{d_3}{f_4}\right)\right) + 1 - \frac{d_3}{f_4}\right)\right) + d_2\left(\frac{-1}{f_3}\left(1 - \frac{d_3}{f_4}\right) - \frac{1}{f_4}\right) + 1 - \frac{d_3}{f_4}\right)$$

Solutions to the above paraxial problem can also be found using damped least-squares optimization with the constraints of having the horizontal and vertical EFLs differ by the anamorphic ratio, while also forcing the BFL to be identical in both axes over a range of focus positions.

An example solution to the paraxial problem is shown below. Referring to the infinity focus arrangement of FIGS. **4**A and **4**C, the following values are assigned:

$f_1 = -100$ mm cylindrical (x-axis)

$f_2 = -240$ mm spherical

$f_3 = 200$ mm cylindrical (x-axis)

$f_4 = 50$ mm spherical

$d_1 = 18$ mm

$d_2 = 37$ mm

$d_3 = 12.7$ mm

The foregoing paraxial combination results in a 50 mm EFL along the y-axis and a 33.4 mm EFL along the x-axis **120**, with a paraxial BFL of 60.4 mm. In this example, the anamorphic ratio at the infinity focus arrangement is 1.5×. When the object **126** moves closer to the anamorphic lens assembly **100** (e.g., to a distance of 500 mm; the distances between the object **126** and the anamorphic lens assembly **100** are not drawn to scale in FIGS. **4**A**4**D), the relative positions of the four lens elements/components **104**, **112**, **114**, **116** are adjusted (e.g., by rotating the focus ring **124**) to form a focused image at the image plane **106**. In this configuration, and referring to the close focus arrangement of FIGS. **4**B and **4**D:

$d_1 = 43.51$ mm

$d_2 = 11.49$ mm

$d_3 = 4.55$ mm

Note that the sum of d_1 and d_2 remains constant as the anamorphic lens assembly **100** is focused, because the distance between the first and second cylindrical lens elements **112**, **114** is fixed. In this paraxial representation, the anamorphic ratio changes slightly (e.g., to 1.468×), and the EFLs along the y-axis and the x-axis, respectively, change to 58.2 mm and 39.7 mm while the paraxial BFL increases to 67.48 mm in the close focus arrangement of FIGS. **4**B and **4**D. Because the astigmatism is corrected by the combination of the three lens elements **112**, **114**, **116**, refocusing by adjusting the primary lens component **104** (moving f_4 relative to f_3), or by moving the image plane **106**, has no impact on image quality.

It should be noted that not all solutions to the above paraxial system are practical, as it is possible to have solutions where one or more of the distances d_1, d_2, or d_3 are negative. Practical solutions may also require that the anamorphic ratio change slightly between the infinity focus arrangement and the close focus arrangement, but in practice it is possible to limit this change in anamorphic ratio to less than a few percent.

TABLE 1 below presents an optical prescription of one example embodiment of the anamorphic lens assembly **100** shown in FIGS. **1**A and **1**B.

TABLE 1

Surface	Radius	Radius	Thickness	Nd	Vd
OBJ	STANDARD	Infinity	600	1176.083	0
1	TOROIDAL	−130.1000	5.0000	1.712995	53.8671
2	TOROIDAL	97.9500	31.5950		
3	STANDARD	−513.0000	3.0000	1.703000	52.3794
4	STANDARD	513.0000	2.9000		
5	TOROIDAL	195.6000	6.0000	1.516330	64.0651
6	TOROIDAL	−107.7000	1.3240		
7	STANDARD	71.5300	4.5000	1.712995	53.8671
8	STANDARD	−1030.0000	0.3000		
9	STANDARD	25.1500	7.0457	1.729157	54.6800
10	STANDARD	28.4500	0.3000		
11	STANDARD	18.8700	4.0000	1.788800	28.4287
12	STANDARD	12.5700	10.5820		
STO	STANDARD	Infinity	5.2520		
14	STANDARD	−12.5200	1.4828	1.788800	28.4287
15	STANDARD	−33.0800	4.8394	1.772499	49.5984
16	STANDARD	−16.8500	0.3000		
17	STANDARD	−302.1000	6.0047	1.740999	52.6365
18	STANDARD	−26.9000	0.3000		
19	STANDARD	51.1000	9.3853	1.589130	61.1350
20	STANDARD	−33.5205	1.5000	1.953749	32.3247
21	STANDARD	−2960.0000	29.9760		
IMA	STANDARD	Infinity	47.6504	0.0000	
THIC 0	INFINITY	600			
THIC 2	8.3000	31.5950			
THIC 4	26.2000	2.9000			
THIC 6	4.8000	1.3240			
THIC 21	26.5000	29.9760			

In some embodiments, the following paraxial solution process may be used to narrow down the range of possible configurations for the anamorphic lens assembly. The paraxial solution process defines the following ten variables:

The focal lengths of the lenses: f_1, f_2, f_3, f_4;

The distances between the lenses at the infinity focus arrangement: d_{1inf}, d_{2inf}, d_{3inf}; and

The distances between the lenses at the close focus arrangement: d_{1close}, d_{2close}, d_{3close}.

Using these ten variables, and based on known and/or desired properties of the resulting anamorphic lens assembly, the paraxial solution process defines the following six equations:

$$d_{1inf} + d_{2inf} = d_{1close} + \quad (1)$$

d_{2close}(due to the fixed spacing of the cylindrical lens elements f_1, f_3)

$$\frac{EFL_{xinf}}{EFL_{yinf}} = \text{the desired value of the} \quad (2)$$

anamorphic ratio at the infinity focus arrangement

$$\frac{EFL_{xclose}}{EFL_{yclose}} = \text{the anamorphic ratio at the close focus arrangement} \quad (3)$$

$$EFL_{xinf} = \text{a constant} \quad (4)$$

(the desired value of *EFL* at the infinity focus arrangement)

$$BFL_{xinf} = \quad (5)$$

BFL_{yinf}(due to the zero astigmatism at the infinity focus arrangement)

$$BFL_{xclose} = \quad (6)$$

BFL_{yclose}(due to the zero astigmatism at the close focus arrangement)

With six equations in ten variables, there are infinite sets of solutions. However, setting two or three of the variables as constants reduces the scope of the solution sets sufficiently to enable solutions to be found using, for example, iterative techniques.

In some embodiments, equations (2) and (3) above will be equal to one another, because the anamorphic ratio will be the same at both the infinity focus arrangement and the close focus arrangement. As discussed above, however, in some embodiments of the present anamorphic lens assemblies the anamorphic ratio may vary slightly between the infinity focus arrangement and the close focus arrangement. For example, the anamorphic ratio may vary by less than 2% between the infinity focus arrangement and the close focus arrangement. In embodiments in which the anamorphic ratio at the infinity focus arrangement is not equal to the anamorphic ratio at the close focus arrangement, equations (2) and (3) above will not be equal to one another. In such embodiments, the lens design process may advantageously attempt to match the anamorphic ratio change of legacy, or classic, anamorphic lenses. In classic anamorphic lenses, the anamorphic ratio is typically smaller at close focus than at infinity focus. There are paraxial solutions that have no change in the anamorphic ratio between close focus and infinity focus, but in practice this condition may not be achievable due to interactions among the principal planes of the various groups of lenses.

As discussed above, in some embodiments one or more of the lens elements may comprise multiple lenses, such as doublets. FIGS. **5**A and **5**B illustrate one such example embodiment. The anamorphic lens assembly **500** of FIGS. **5**A and **5**B includes a first cylindrical lens element **502** and a second cylindrical lens element **504** that each comprise doublets. The doublet of the first cylindrical lens element **502** comprises two negative cylinders, while the doublet of the second cylindrical lens element **504** comprises two cylinders that together have positive power and reduce color aberrations. In some embodiments, the doublet of the second cylindrical lens element **504** may comprise two positive cylinders. The first spherical lens element **506** has negative refractive power, such that the first spherical lens element **506** moves toward the primary lens component **508** as the anamorphic lens assembly **500** transitions away from the

infinity focus arrangement of FIG. **5**A and toward the close focus arrangement of FIG. **5**B. In the illustrated embodiment, the anamorphic ratio of the anamorphic lens assembly **500** may comprise, for example, a 1.8× squeeze (or any other squeeze).

As discussed above, in some embodiments the first spherical lens element may have positive refractive power. FIGS. **6**A and **6**B illustrate one such example embodiment. The anamorphic lens assembly **600** of FIGS. **6**A and **6**B includes a first spherical lens element **602** comprising a doublet with positive refractive power. The first spherical lens element **602** moves toward the primary lens component **604** as the anamorphic lens assembly **600** transitions away from the infinity focus arrangement of FIG. **6**A and toward the close focus arrangement of FIG. **6**B.

In the paraxial representation of some embodiments, the distance d_3 may not change as the focus of the anamorphic lens assembly is adjusted. Instead, the BFL (the distance between f_4 and the image plane **106**) may change. FIGS. **7**A and **7**B illustrate an example embodiment of one such anamorphic lens assembly **700**. FIG. **7**A illustrates the anamorphic lens assembly **700** in the infinity focus arrangement, and FIG. **7**B illustrates the anamorphic lens assembly **700** in the close focus arrangement. As the anamorphic lens assembly **700** transitions from the infinity focus arrangement (FIG. **7**A) to the close focus arrangement (FIG. **7**B), d_1 increases while d_2 decreases such that the sum of d_1 and d_2 remains constant. d_3 remains constant while the BFL of the anamorphic lens assembly **700** increases. In alternative embodiments, the relative changes in d_1, d_2, d_3, and BFL may vary, particularly as the focal lengths of the various lenses are varied. For example, if the first spherical lens element **116** in the example shown in FIGS. **7**A and **7**B has a negative focal length f_2, then in an alternative embodiment in which the first spherical lens element **116** has a positive focal length f_2, as the anamorphic lens assembly transitions from the infinity focus arrangement to the close focus arrangement d_1 may decrease while d_2 increases, d_3 remains constant, and the BFL of the anamorphic lens assembly decreases. Further, paraxial solutions exist for embodiments in which d_3 changes and the BFL remains constant, and the focusing mechanism of the primary lens component **104** doesn't necessarily require moving the primary lens component **104** away from the image plane **106**. For example, these conditions may be satisfied in embodiments in which the primary lens component **104** includes an internal focus mechanism in which internal lens elements of the primary lens component **104** are movable, but the primary lens component **104** as a whole does not move relative to the anamorphic lens component **102**.

FIGS. **8**A and **8**B illustrate another anamorphic lens assembly **800** according to some embodiments. Similar to the embodiment of FIGS. **1**A and **1**B, the anamorphic lens assembly **800** includes an anamorphic lens component **802**, a primary lens component **804**, a first cylindrical lens element **812**, a second cylindrical lens element **814**, and a first spherical lens element **816** disposed between the first and second cylindrical lens elements **812**, **814**. Relative positions of the first and second cylindrical lens elements **812**, **814** are fixed along an optical axis **818** of the anamorphic lens assembly **800**, while the first spherical lens element **816** is translatable along the optical axis **818** with respect to the first and second cylindrical lens elements **812**, **814**. TABLE 2 below presents an optical prescription of the example embodiment of the anamorphic lens assembly **800** shown in FIGS. **8**A and **8**B.

TABLE 2

Surface	Radius	Thickness	Nd	Vd
OBJ	Infinity	Infinity		
1	−183.8333	5.0000	1.729157	54.68001
2	95.0298	7.0000		
3	200.0000	3.0000	1.516330	64.06513
4	91.6639	33.5000		
5	201.1614	6.0000	1.528410	76.45282
6	−139.0105	5.1260		
7	173.0229	2.5000	1.712995	53.86706
8	−291.4164	0.3000		
9	34.8983	8.1731	1.804000	46.52753
10	−44.5651	2.3587	1.720467	34.70798
11	42.4306	6.0000		
STO	Infinity	5.6507		
13	−18.6601	1.3000	1.720002	46.02457
14	−163.3591	2.0000		
15	−30.2437	6.8506	1.793600	37.08945
16	−21.5280	0.3000		
17	99.9762	4.0000	1.743198	49.33944
18	−75.9795	0.3000		
19	77.6224	5.7235	1.804000	46.52753
20	−46.2051	2.1386		
21	−37.1102	1.5000	1.959060	17.47125
22	−600.5993	28.3926		
IMA	Infinity			
THIC 0		Infinity	600.0000	
THIC 2		7.0000	31.2100	
THIC 4		33.5000	9.2900	
THIC 6		5.1260	1.4000	
THIC 22		28.3926	32.1180	

In some embodiments, the spherical lens element of the anamorphic lens component may comprise multiple lenses, such as two lenses, and the primary lens component may comprise internal focus. FIGS. **9**A and **9**B illustrate one such example embodiment. The anamorphic lens assembly **900** of FIGS. **9**A and **9**B includes an anamorphic lens component **902** and a primary lens component **904**. In FIG. **9**A the anamorphic lens assembly **900** is shown in an infinity focus arrangement, while in FIG. **9**B the anamorphic lens assembly **900** is shown in a close focus arrangement.

With reference to FIG. **9**A, the anamorphic lens component **902** includes a first cylindrical lens element **912**, a second cylindrical lens element **914**, and first and second spherical lens elements **916**(**1**), **916**(**2**) disposed between the first and second cylindrical lens elements **912**, **914**. In the illustrated embodiment, the first cylindrical lens element **912** comprises a singlet with negative refractive power (in the horizontal direction), and the second cylindrical lens element **914** comprises a doublet including a negative refractive power (in the horizontal direction) cylindrical lens **914**(**1**) and a positive refractive power (in the horizontal direction) cylindrical lens **914**(**2**). In some embodiments, the combined refractive powers of the lenses **914**(**1**), **914**(**2**) is positive, and the combined refractive power of the first and second cylindrical lens elements **912**, **914** provides a desired increase or decrease in the beam diameter in the horizontal direction (e.g., 2×, or 0.5×, or any other ratio), as described above with respect to previous embodiments. In alternative embodiments, either or both of the first and second cylindrical lens elements **912**, **914** may have different configurations, such as any of the configurations shown in other embodiments herein.

Relative positions of the first and second cylindrical lens elements **912**, **914** are fixed along an optical axis **918** of the anamorphic lens assembly **900**, while the first and second spherical lens elements **916**(**1**), **916**(**2**) are translatable along the optical axis **918** with respect to the first and second cylindrical lens elements **912**, **914**, as shown in FIGS. **9**A

and **9**B. In some embodiments, the fixed distance between the first and second cylindrical lens elements **912**, **914**, coupled with the ability of the first and second spherical lens elements **916**(**1**), **916**(**2**) to translate along the optical axis **918** between the first and second cylindrical lens elements **912**, **914**, advantageously allows the spacings between the first and second spherical lens elements **916**(**1**), **916**(**2**) and the first and second cylindrical lens elements **912**, **914**, respectively, to be adjustable. This feature contributes to the advantageous optical characteristics of some of the present embodiments, as described above with respect to previous embodiments.

With further reference to FIGS. **9**A and **9**B, the first and second spherical lens elements **916**(**1**), **916**(**2**) are translatable along the optical axis **918** with respect to each other, such that a spacing between the first and second spherical lens elements **916**(**1**), **916**(**2**) is variable as the first and second spherical lens elements **916**(**1**), **916**(**2**) translate along the optical axis **918**. In the illustrated embodiment, the spacing between the first and second spherical lens elements **916**(**1**), **916**(**2**) increases as the anamorphic lens assembly **900** transitions away from the infinity focus arrangement of FIG. **9**A toward the close focus arrangement of FIG. **9**B. In alternative embodiments, this configuration may be reversed such that the spacing between the first and second spherical lens elements **916**(**1**), **916**(**2**) decreases as the anamorphic lens assembly **900** transitions away from the infinity focus arrangement of FIG. **9**A toward the close focus arrangement of FIG. **9**B. In further alternative embodiments, the first and second spherical lens elements **916**(**1**), **916**(**2**) may move in the opposite direction (e.g., away from the first cylindrical lens element **912**) when transitioning away from the infinity focus arrangement toward the close focus arrangement, and in such embodiments the spacing between the first and second spherical lens elements **916**(**1**), **916**(**2**) may increase or decrease when transitioning away from the infinity focus arrangement toward the close focus arrangement. In still further alternative embodiments, the spacing between the first and second spherical lens elements **916**(**1**), **916**(**2**) may be fixed, such that the spacing between the lens elements **916**(**1**), **916**(**2**) doesn't change as the lens elements **916**(**1**), **916**(**2**) translate along the optical axis **918**.

In some embodiments, the anamorphic lens assembly **900** of FIGS. **9**A and **9**B may have a long focal length and a large entrance pupil. For example, the focal length may be in the range of 90 mm to 150 mm, and a diameter of the entrance pupil may be in the range of 35 mm to 60 mm. In such embodiments, the second spherical lens element **916**(**2**) may advantageously compensate for any spherical aberration created by the first spherical lens element **916**(**1**). For example, in some embodiments the second spherical lens element **916**(**2**) may comprise a meniscus lens having two spherical curved surfaces, convex on a first side (the object side **920**) and concave on a second side (the image side **922**), and a greater thickness at the center than at the edges (due to the convex side **920** having a smaller radius of curvature than the concave side **922**). In general, meniscus lenses provide a smaller beam diameter to reduce beam waist and spherical aberration. When a meniscus lens (e.g., the second spherical lens element **916**(**2**)) is combined with another lens (e.g., the first spherical lens element **916**(**1**)), the focal length is shortened and the numerical aperture of the system is increased. This advantageously reduces the image distortion and increases the image resolution. In the illustrated embodiment of FIGS. **9**A and **9**B, the first and second spherical lens elements **916**(**1**), **916**(**2**) in combination have more surfaces (as compared to single-spherical lens element embodiments) to create less spherical aberration, and the meniscus lens **916**(**2**) compensates for the spherical aberration created by the first spherical lens element **916**(**1**). In particular, FIGS. **10**A and **10**B are ray aberration plots on-axis at infinity focus and close focus, respectively, for the anamorphic lens assembly **900** of FIGS. **9**A and **9**B. These plots illustrate that the spherical aberration (plotted on the vertical axis in FIGS. **10**A and **10**B) caused by the first and second spherical lens elements **916**(**1**), **916**(**2**) changes dramatically as the lens elements **916**(**1**), **916**(**2**) move along the optical axis **918** (plotted on the horizontal axis in FIGS. **10**A and **10**B). In general, spherical aberration can be reduced by splitting the optical power between two lens elements, or, in some cases, adding a lens element to create the opposite spherical aberration can improve performance. Since spherical aberration is a non-linear function of the aperture diameter, using multiple elements as the compensating group (e.g., the first and second spherical lens elements **916**(**1**), **916**(**2**)) is advantageous on a long focal length lens.

With reference to FIG. **9**A, the primary lens component **904** is disposed on a side of the second cylindrical lens element **914** opposite the first and second spherical lens elements **916**(**1**), **916**(**2**), and includes one or more primary spherical lens elements **926**. The primary spherical lens elements **926** may be configured to provide primary imaging by, for example, varying a focus of an image formed by the primary lens component **904**. In the illustrated embodiment, primary imaging is provided by internal focus of the primary lens component **904**. In particular, a first one of the primary spherical lens elements **926**(**1**) is translatable along the optical axis **918** with respect to the other primary spherical lens elements **926**. In contrast to earlier embodiments in which the primary lens component **104** is movable with respect to the anamorphic lens component **102**, in the embodiment of FIGS. **9**A and **9**B only the first primary spherical lens element **926**(**1**) is movable with respect to the anamorphic lens component **902**. The positions of all other primary spherical lens elements **926** (besides the first primary spherical lens element **926**(**1**)) are fixed relative to the anamorphic lens component **902**.

The embodiment of FIGS. **9**A and **9**B, in which the primary lens component **904** includes an internal focus mechanism, advantageously reduces the focal length of the primary lens component **904** at close focus. In embodiments described above that use external focus (e.g., FIGS. **1**-**8**), the focal length of the anamorphic lens assembly increases at close focus. This property can cause breathing, which is the tendency for an object image to increase in size as the lens focal distance is reduced. The primary lens component **904** in the embodiment of FIGS. **9**A and **9**B has a reduced focal length at close focus, which offsets the focal length increase of the anamorphic lens assembly **900** and reduces the amount of breathing that occurs at close focus. In some embodiments the focal length of the anamorphic lens assembly **900** stays relatively constant between the infinity focus arrangement (FIG. **9**A) and the close focus arrangement (FIG. **9**B). For example, in some embodiments the focal length of the anamorphic lens assembly **900** may vary by less than 10%, such as less than 5%, or less than 4%, or less than 3%, or less than 2%, or less than 1%. Also in some embodiments the anamorphic ratio of the anamorphic lens assembly **900** may stay relatively constant between the infinity focus arrangement (FIG. **9**A) and the close focus arrangement (FIG. **9**B), varying by less than 10%, such as less than 5%, or less than 4%, or less than 3%, or less than 2%, or less than 1%.

In the illustrated embodiment, the first primary spherical lens element **926**(**1**) comprises a doublet having a positive refractive power. In alternative embodiments, the first primary spherical lens element **926**(**1**) may have a different configuration, such as a singlet and/or a positive refractive power. In some embodiments, the primary spherical lens elements **926** may be configured to vary other optical properties of the image, such as a softness of the image, a size of the image, or may be configured to correct for blur or aberrations in the image, or to change other optical properties as desired. While the primary spherical lens elements **926** are described herein as being spherical, in some embodiments one or more of the primary spherical lens elements **926** may be aspherical if desired.

With reference to FIG. **9**A, and as described above, the relative positions of the first and second cylindrical lens elements **912**, **914** are fixed along the optical axis **918** of the anamorphic lens component **902**, and the relative positions of the primary spherical lens elements **926** (except the first primary spherical lens element **926**(**1**)) are fixed along the optical axis **918** of the primary lens component **904**. The first and second spherical lens elements **916**(**1**), **916**(**2**) are, however, translatable along the optical axis **918** with respect to the first and second cylindrical lens elements **912**, **914**, and the first primary spherical lens element **926**(**1**) is translatable along the optical axis **918** with respect to the other primary spherical lens elements **926**. For example, FIG. **9**A illustrates the anamorphic lens assembly **900** in an infinity focus arrangement in which the first and second spherical lens elements **916**(**1**), **916**(**2**) are disposed at the limit of their travel in the direction toward the second cylindrical lens element **914** and the first primary spherical lens element **926**(**1**) is disposed at the limit of its travel in the direction toward the anamorphic lens component **902**, while FIG. **9**B illustrates the anamorphic lens assembly **900** in a close focus arrangement in which the first and second spherical lens elements **916**(**1**), **916**(**2**) are disposed at the limit of their travel in the direction toward the first cylindrical lens element **912** and the first primary spherical lens element **926**(**1**) is disposed at the limit of its travel in the direction away from the anamorphic lens component **902**.

As discussed above, the spacing between the first and second spherical lens elements **916**(**1**), **916**(**2**) may be variable as the first and second spherical lens elements **916**(**1**), **916**(**2**) translate along the optical axis **918**. In some embodiments, this variable spacing may be accomplished by positioning the lens elements **916**(**1**), **916**(**2**) in separate cells (e.g., rings that hold the individual lens elements) with individual cam slots to determine the respective travel paths of each lens element **916**(**1**), **916**(**2**). For example, movements of the first and second spherical lens elements **916**(**1**), **916**(**2**) may be controlled by respective cams (not shown) that rotate with a focus ring **924** disposed about the primary lens component **904**. In some embodiments, rotation of the focus ring **924** about the primary lens component **904** controls focus of the anamorphic lens assembly **900** by adjusting a position of the first primary spherical lens element **926**(**1**) with respect to the other primary spherical lens elements **926**, and by adjusting the positions of the first and second spherical lens elements **916**(**1**), **916**(**2**) with respect to the first and second cylindrical lens elements **912**, **914**. That is, rotation of the focus ring **924** about the primary lens component **904** controls movement of each of the first primary spherical lens element **926**(**1**) and the first and second spherical lens elements **916**(**1**), **916**(**2**), with each of the lens element **926**(**1**), **916**(**1**), **916**(**2**) having a respective cam that moves in a respective cam slot as the focus ring **924** rotates. In some embodiments, the cam slots may be helical, and may have different pitches so that the lens elements **926**(**1**), **916**(**1**), **916**(**2**) translate along the optical axis **918** at different rates as the focus ring **924** rotates. Also as discussed above, the first and second spherical lens elements **916**(**1**), **916**(**2**) may be fixed with respect to one another. In such embodiments, the cam slots corresponding to the first and second spherical lens elements **916**(**1**), **916**(**2**) may have the same pitch, or the cams corresponding to the first and second spherical lens elements **916**(**1**), **916**(**2**) may be disposed in the same cam slot.

Similar to embodiments described above, movement relationships between (or among) the focus ring **924** and one or more of the first spherical lens element **916**(**1**), the second spherical lens element **916**(**2**), and/or the first primary spherical lens element **926**(**1**) may be defined by mathematical relationships. For example, in some embodiments movement of the first spherical lens element **916**(**1**) relative to the focus ring is defined by a first mathematical relationship, and movement of the second spherical lens element **916**(**2**) relative to the focus ring is defined by a second mathematical relationship (where the first and second mathematical relationships are different). In embodiments in which the spacing between the first and second spherical lens elements **916**(**1**), **916**(**2**) is variable, movement of the first spherical lens element relative to the second spherical lens element may be defined by another mathematical relationship. In some embodiments, movement of the first and second spherical lens elements relative to the first one of the primary spherical lens elements is defined by a still further mathematical relationship. Any of the foregoing mathematical relationships may comprise, for example, 2nd or 3rd order polynomials, and may be defined using the paraxial solution process described above for every focal distance. Alternatively, a computer simulation may be used to find solutions at multiple focal distances, and then a least-squares curve fit may be used to determine the polynomial coefficients.

TABLE 3 below presents an optical prescription of one example embodiment of the anamorphic lens assembly **900** shown in FIGS. **9**A and **9**B.

TABLE 3

Type		Radius	Thickness	Nd	Vd
OBJ	STANDARD	Infinity	Infinity	AIR	
1	TOROIDAL	−202.0114	5.0000	1.6400	60.0781
2	TOROIDAL	183.1334	25.3000	AIR	
3	STANDARD	90.0829	4.0000	1.8044	39.5862
4	STANDARD	74.5503	5.7900	AIR	
5	STANDARD	93.9045	7.0000	1.6180	63.3335
6	STANDARD	623.8520	22.1200	AIR	
7	STANDARD	949.7368	3.0000	1.6140	54.9711
8	STANDARD	125.0782	3.0000	AIR	
9	TOROIDAL	211.8012	6.0000	1.5952	67.7357

TABLE 3-continued

10	TOROIDAL	−193.1621	0.5000	AIR	
11	STANDARD	121.9602	6.0000	1.4875	70.2103
12	STANDARD	577.8786	0.5000	AIR	
13	STANDARD	78.1681	12.0000	1.6129	37.0053
14	STANDARD	−66.7128	2.0000	1.6541	39.6828
15	STANDARD	199.8516	3.0000	AIR	
16	STANDARD	−441.7160	3.5000	1.8467	23.7779
17	STANDARD	216.9168	2.5000	AIR	
18	STANDARD	64.8562	7.7000	1.4970	81.5459
19	STANDARD	−193.9164	0.5000	AIR	
20	STANDARD	25.6559	3.2000	1.7215	29.2323
21	STANDARD	22.5245	12.5000	AIR	
STO	STANDARD	Infinity	0.1500	AIR	
23	STANDARD	138.3642	4.5000	1.8081	22.7608
24	STANDARD	−1075.7460	2.0000	1.7340	51.4706
25	STANDARD	42.8231	24.6500	AIR	
26	STANDARD	−26.4274	2.5000	1.7215	29.2323
27	STANDARD	150.2071	8.5000	1.7639	48.4887
28	STANDARD	−33.1138	0.5000	AIR	
29	STANDARD	94.2970	6.0000	1.7859	44.2026
30	STANDARD	−466.4518	53.5000	AIR	
31	STANDARD	Infinity	2.7300	AIR	
IMA	STANDARD	Infinity		0.0000	

Configuration	THIC 0	THIC 2	THIC 4	THIC 6	THIC 22	THIC 25
Infinity	Infinity	25.3160	5.7950	22.1200	0.1500	24.65
Close	900	3.1580	8.0500	42.0300	15.0000	9.8

FIGS. **11**A and **11**B are schematic diagrams illustrating another anamorphic lens assembly according to some embodiments. The anamorphic lens assembly **1100** of FIGS. **11**A and **11**B includes an anamorphic lens component **1102** and a primary lens component **1104**. In FIG. **11**A the anamorphic lens assembly **1100** is shown in an infinity focus arrangement, while in FIG. **11**B the anamorphic lens assembly **1100** is shown in a close focus arrangement.

With reference to FIG. **11**A, the anamorphic lens component **1102** includes a first cylindrical lens element **1112**, a second cylindrical lens element **1114**, and a spherical lens element **1116** disposed between the first and second cylindrical lens elements **1112**, **1114**. In the illustrated embodiment, the first cylindrical lens element **1112** comprises a cylindrical lens **1112**(**1**) with negative refractive power (in the horizontal direction) and a negative spherical lens **1112**(**2**), and the second cylindrical lens element **1114** comprises a doublet including a positive refractive power (in the horizontal direction) cylindrical lens **1114**(**1**) and a negative refractive power (in the horizontal direction) cylindrical lens **1114**(**2**). In some embodiments, the negative refractive power of the cylindrical lens **1114**(**1**) is relatively weak in comparison to the positive refractive power of the cylindrical lens **1114**(**2**), such that the combined refractive powers of the lenses **1114**(**1**), **1114**(**2**) is positive. Also in some embodiments, the cylindrical lens **1114**(**2**) can be either a positive or negative element, but weak in optical power compared to the cylindrical lens **1114**(**1**). Also in some embodiments, the combined refractive power of the first and second cylindrical lens elements **1112**, **1114** provides a desired increase or decrease in the beam diameter in the horizontal direction (e.g., 2×, or 0.5×, or any other ratio), as described above with respect to previous embodiments. In alternative embodiments, either or both of the first and second cylindrical lens elements **1112**, **1114** may have different configurations, such as any of the configurations shown in other embodiments herein. For example, the first cylindrical lens element **1112** may comprise two negative cylinders close together instead of a single cylinder.

Relative positions of the first cylindrical lens element **1112** and the cylindrical lens **1114**(**1**) of the second cylindrical lens element **1114** are fixed along an optical axis **1118** of the anamorphic lens assembly **1100**, while the cylindrical lens **1114**(**2**) of the second cylindrical lens element **1114** is translatable along the optical axis **1118** with respect to both the first cylindrical lens element **1112** and the cylindrical lens **1114**(**1**) of the second cylindrical lens element **1114**, as shown in FIGS. **11**A and **11**B. In the illustrated embodiment, the cylindrical lens **1114**(**2**) of the second cylindrical lens element **1114** moves with the primary lens component **1104** as the anamorphic lens assembly **1100** transitions between the infinity focus arrangement (FIG. **11**A) and the close focus arrangement (FIG. **11**B). Thus, the combined power of the cylindrical lenses **1114**(**1**), **1114**(**2**) changes as the distance between them changes. This property advantageously reduces the change in the anamorphic ratio of the overall anamorphic lens assembly **1100** between the infinity focus arrangement (FIG. **11**A) and the close focus arrangement (FIG. **111**B), because the change in anamorphic ratio that results from movement of the spherical lens element **1116** is offset by the change in the combined power of the cylindrical lenses **1114**(**1**), **1114**(**2**). The amount of astigmatism that needs to be corrected is also advantageously reduced, such that the spherical lens element **1116** doesn't have to move as far, and the overall change in focal length of the anamorphic lens assembly **1100** is reduced.

In the preceding description, various embodiments are described. For purposes of explanation, specific configurations and details are set forth in order to provide a thorough understanding of the embodiments. However, it will also be apparent to one skilled in the art that the embodiments can be practiced without the specific details. Furthermore, well-known features can be omitted or simplified in order not to obscure the embodiment being described.

References to "one embodiment," "an embodiment," "an example embodiment," etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic.

Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

Moreover, in the various embodiments described above, unless specifically noted otherwise, disjunctive language such as the phrase "at least one of A, B, or C" is intended to be understood to mean any of A, B, or C, or any combination thereof (e.g., A, B, and/or C). Similarly, language such as "at least one or more of A, B, and C" (or "one or more of A, B, and C") is intended to be understood to mean any of A, B, or C, or any combination thereof (e.g., A, B, and/or C). As such, disjunctive language is not intended to, nor should it be understood to, imply that a given embodiment requires at least one of A, at least one of B, and at least one of C to each be present.

As used herein, the term "based on" (or similar) is an open-ended term used to describe one or more factors that affect a determination or other action. This term does not foreclose additional factors that may affect a determination or action. For example, a determination may be solely based on the factor(s) listed or based on the factor(s) and one or more additional factors. Thus, if an action A is "based on" B, then B is one factor that affects action A, but this does not foreclose the action A from also being based on one or more other factors, such as factor C. However, in some instances, action A may be based entirely on B.

Unless otherwise explicitly stated, articles such as "a" or "an" should generally be interpreted to include one or multiple described items. Accordingly, phrases such as "a device configured to" or "a computing device" are intended to include one or multiple recited devices. Such one or more recited devices can be collectively configured to carry out the stated operations. For example, "a processor configured to carry out operations A, B, and C" can include a first processor configured to carry out operation A working in conjunction with a second processor configured to carry out operations B and C.

Further, the words "may" or "can" are used in a permissive sense (meaning having the potential to), rather than the mandatory sense (meaning must). The words "include," "including," and "includes" are used to indicate open-ended relationships and therefore mean including, but not limited to. Similarly, the words "have," "having," and "has" also indicate open-ended relationships, and thus mean having, but not limited to. The terms "first," "second," "third," and so forth as used herein are used as labels for the nouns that they precede, and do not imply any type of ordering (e.g., spatial, temporal, logical, etc.) unless such an ordering is otherwise explicitly indicated.

The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that various modifications and changes can be made thereunto without departing from the broader scope of the disclosure as set forth in the claims.

What is claimed is:

1. An anamorphic lens assembly for a camera, the anamorphic lens assembly comprising:

an anamorphic lens component; and

a primary lens component;

the anamorphic lens component comprising:

a negative power first cylindrical lens element;

a positive power second cylindrical lens element, relative positions of the first and second cylindrical lens elements being fixed along an optical axis of the anamorphic lens assembly;

a first spherical lens element disposed between the first and second cylindrical lens elements and translatable along the optical axis with respect to the first and second cylindrical lens elements; and

a second spherical lens element disposed between the first and second cylindrical lens elements and translatable along the optical axis with respect to the first and second cylindrical lens elements, a spacing between the first and second spherical lens elements being variable as the first and second spherical lens elements translate along the optical axis;

the primary lens component comprising:

one or more primary spherical lens elements disposed on a side of the second cylindrical lens element opposite the first and second spherical lens elements;

an order of the lens elements comprising, from an object side toward an image side: the first cylindrical lens element, the first and second spherical lens elements, the second cylindrical lens element, and the one or more primary spherical lens elements;

optical characteristics of the first cylindrical lens element, the first and second spherical lens elements, and the second cylindrical lens element in combination producing zero astigmatism in an image plane for an object at infinity focus;

the optical characteristics of the first cylindrical lens element, the first and second spherical lens elements, and the second cylindrical lens element in combination being adjustable by the first and second spherical lens elements translating along the optical axis with respect to the first and second cylindrical lens elements, such that the first cylindrical lens element, the first and second spherical lens elements, and the second cylindrical lens element in combination produce a first astigmatism that is opposite to a second astigmatism produced by unequal movement of the image plane along horizontal and vertical axes as the object moves from the infinity focus toward the anamorphic lens assembly; and

optical characteristics of the primary lens component being adjustable by at least a first one of the primary spherical lens elements translating along the optical axis with respect to at least a second one of the primary spherical lens elements.

2. The anamorphic lens assembly of claim **1**, the primary lens component further comprising a focus ring, wherein rotation of the focus ring about the primary lens component causes translation of the first and second spherical lens elements along the optical axis.

3. The anamorphic lens assembly of claim **2**, wherein movement of the first spherical lens element relative to the focus ring is defined by a first mathematical relationship, and wherein movement of the second spherical lens element relative to the focus ring is defined by a second mathematical relationship that is different from the first mathematical relationship.

4. An anamorphic lens assembly comprising:

a first cylindrical lens element;

a second cylindrical lens element;

a first spherical lens element disposed between the first and second cylindrical lens elements and moveable with respect to the first and second cylindrical lens elements; and

a second spherical lens element disposed between the first and second cylindrical lens elements and moveable with respect to the first and second cylindrical lens elements;

the optical characteristics of the first cylindrical lens element, the first and second spherical lens elements, and the second cylindrical lens element in combination being adjustable as the first and second spherical lens elements move with respect to the first and second cylindrical lens elements such that the first cylindrical lens element, the first and second spherical lens elements, and the second cylindrical lens element in combination produce a first astigmatism that is opposite to a second astigmatism produced by unequal movement of an image plane along horizontal and vertical axes as an object moves from infinity focus toward the anamorphic lens assembly.

5. The anamorphic lens assembly of claim **4**, wherein a spacing between the first and second spherical lens elements is variable as the first and second spherical lens elements move with respect to the first and second cylindrical lens elements.

6. The anamorphic lens assembly of claim **5**, further comprising a primary lens component including a focus adjustment member disposed around the primary lens component.

7. The anamorphic lens assembly of claim **6**, wherein movement of the focus adjustment member about the primary lens component causes movement of the first and second spherical lens elements relative to one another.

8. The anamorphic lens assembly of claim **7**, wherein movement of the first spherical lens element relative to the focus adjustment member is defined by a first mathematical relationship, and wherein movement of the second spherical lens element relative to the focus adjustment member is defined by a second mathematical relationship.

9. The anamorphic lens assembly of claim **6**, wherein the focus adjustment member, the primary lens component, and the first and second spherical lens elements are mechanically coupled such that as the focus adjustment member is rotated in a first rotational direction around the primary lens component the primary lens component is adjusted away from an infinity focus arrangement and toward a close focus arrangement while the first and second spherical lens elements travel in one of:

a first direction toward the second cylindrical lens element and away from the first cylindrical lens element; or

a second direction away from the second cylindrical lens element and toward the first cylindrical lens element.

10. The anamorphic lens assembly of claim **9**, wherein as the focus adjustment member is rotated in the first rotational direction around the primary lens component a first one of the primary spherical lens elements travels in one of:

a first direction away from the first and second cylindrical lens elements; or

a second direction toward the first and second cylindrical lens elements.

11. The anamorphic lens assembly of claim **4**, wherein optical characteristics of the first cylindrical lens element, the first and second spherical lens elements, and the second cylindrical lens element in combination produce zero astigmatism in the image plane for the object at infinity focus.

12. The anamorphic lens assembly of claim **11**, wherein changes to the focal lengths cause an anamorphic ratio of the anamorphic lens assembly to also change.

13. The anamorphic lens assembly of claim **12**, wherein a difference in the anamorphic ratio is less than 2% between the infinity focus arrangement and the close focus arrangement.

14. An anamorphic lens assembly for a camera, the anamorphic lens assembly comprising:

an anamorphic lens component; and

a primary lens component;

the anamorphic lens component comprising:

a negative power first cylindrical lens element;

a negative power second cylindrical lens element, relative positions of the first and second cylindrical lens elements being fixed along an optical axis of the anamorphic lens assembly;

a positive power third cylindrical lens element, the third cylindrical lens element being translatable along the optical axis with respect to the first and second cylindrical lens elements; and

a first spherical lens element disposed between the first and second cylindrical lens elements and translatable along the optical axis with respect to the first and second cylindrical lens elements;

the primary lens component comprising:

one or more primary spherical lens elements disposed on a side of the third cylindrical lens element opposite the first spherical lens element;

an order of the lens elements comprising, from an object side toward an image side: the first cylindrical lens element, the first spherical lens element, the second cylindrical lens element, the third cylindrical lens element, and the one or more primary spherical lens elements;

optical characteristics of the first cylindrical lens element, the first spherical lens element, the second cylindrical lens element, and the third cylindrical lens element in combination producing zero astigmatism in an image plane for an object at infinity focus; and

the optical characteristics of the first cylindrical lens element, the first spherical lens element, the second cylindrical lens element, and the third cylindrical lens element in combination being adjustable by the first spherical lens element translating along the optical axis with respect to the first, second, and third cylindrical lens elements, such that the first cylindrical lens element, the first spherical lens element, the second cylindrical lens element, and the third cylindrical lens element in combination produce a first astigmatism that is opposite to a second astigmatism produced by unequal movement of the image plane along horizontal and vertical axes as the object moves from the infinity focus toward the anamorphic lens assembly.

15. The anamorphic lens assembly of claim **14**, the primary lens component further comprising a focus adjustment member disposed around the primary lens component, wherein movement of the focus adjustment member about the primary lens component causes translation of the third cylindrical lens element along the optical axis with respect to the first and second cylindrical lens elements.

16. The anamorphic lens assembly of claim **15**, wherein movement of the third cylindrical lens element relative to the focus adjustment member is defined by a mathematical relationship.

17. The anamorphic lens assembly of claim **15**, wherein the focus adjustment member, the primary lens component, the first spherical lens element, and the third cylindrical lens element are mechanically coupled such that as the focus adjustment member is rotated in a first rotational direction

around the primary lens component the primary lens component is adjusted away from an infinity focus arrangement and toward a close focus arrangement while the first spherical lens element and the third cylindrical lens element travel in one of:

a first direction toward the first cylindrical lens element; or

a second direction away from the first cylindrical lens element.

18. The anamorphic lens assembly of claim **14**, wherein optical characteristics of the first cylindrical lens element, the first spherical lens element, the second cylindrical lens element, and the third cylindrical lens element in combination produce zero astigmatism in the image plane for the object at infinity focus.

19. The anamorphic lens assembly of claim **14**, wherein as the anamorphic lens assembly transitions between an infinity focus arrangement and a close focus arrangement, horizontal and vertical focal lengths of a combination of the first cylindrical lens element, the first spherical lens element, the second cylindrical lens element, and the third cylindrical lens element change due to changes in a first spacing between the first cylindrical lens element and the first spherical lens element and a second spacing between the second cylindrical lens element and the first spherical lens element.

20. The anamorphic lens assembly of claim **19**, wherein changes to the focal lengths cause an anamorphic ratio of the anamorphic lens assembly to also change.

* * * * *